
6G-EWOC: Crowdsourced SLAM data fusion for Safe and Efficient ADAS Driving

Item Type Conference Paper

Author J. A. Lázaro

Author J. R. Casas

Author J. Ruiz-Hidalgo

Author M. C. Garcia

Author G. M. Pey

Author J. S. Pujol

Author M. Wang

Author E. Theodoropoulou

Author G. Lyberopoulos

Author C. Marcus

Author O. Eriksson

Author P. García

Author S. Royo

Author J. Riu

Author B. Schrenk

Author J. M. Fàbrega

Abstract The development of transport infrastructures for Advanced Driver Assistance Systems (ADAS) and autonomous vehicles operating efficiently and safely in congestion-free traffic flows is a major challenge for telecommunications technologies. Simultaneous Localization and Mapping (SLAM) plays a crucial role in ensuring uninterrupted journeys for emergency vehicles and increasing the safety of vulnerable road users in complex traffic scenarios. Accurate SLAM mapping for ADAS systems requires data from different sensor technologies – such as high-resolution cameras or Radio/Light Detection and Ranging (RaDAR/LiDAR)– to be effectively combined or fused. Sensor fusion results in high data throughput and low latency requirements. However, optimal mapping outcomes occur when processing systems fuse data from sensors positioned at diverse locations within the traffic scene. By crowdsourcing diverse sensors, we can multiply the view angles, mitigate occlusions and improve the overall scene coverage. Yet, this approach introduces additional challenges for communication systems within both the vehicles and the infrastructure. Addressing these challenges is essential for seamless development of safe and efficient ADAS driving techniques.

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Tags:

Autonomous vehicles, Safety, Sensor fusion, Data integration, Autonomous Driving, Roads, Telecommunications, 6G, 6G mobile communication, Three-dimensional displays, 3D Object Detection, 3D Scene Completion, Crowdsourced SLAM, Data Fusion, Free-space optical communication, Optical communication terminals, Simultaneous localization and mapping, Throughput

6G-EWOC: Optical Wireless Communication in Support of Autonomous Driving

Item Type Conference Paper

Author B. Schrenk

Author E. Theodoropoulou

Author G. Lyberopoulos

Author C. Marcus

Author O. Eriksson

Author J. A. Lázaro

Author J. R. Casas

Author F. D. Otin

Author P. García

Author S. Royo

Author J. Riu

Author J. M. Fàbrega

Abstract The road towards an efficient transport infrastructure, which safely operates fully autonomous vehicles in congestion-free traffic flows, is subject to several technological challenges that reach also into the realm of telecommunications. Future 6G networks are required to handle huge amounts of sensor information, both in terms of low-latency communication and real-time data processing. The 6G-EWOC project contributes to this challenge through a concerted use of optical wireless communication and sensing with radio-frequency based cellular technology. Three optical-wireless scenarios are discussed, including (i) instant data exchange between vehicles leveraging their lighting assets for low-latency communication, (ii) the provision of vehicular data off-loading to a fiber-based 6G infrastructure through beamsteering-assisted optical remote radio heads to enable data fusion at the edge cloud, and (iii) the seamless wireless extension of fiber-grade capacity to isolated network segments situated in fiber-scarce brownfields, which is accomplished through transparent optical air interfaces that ensure fiber-grade continuity by preventing a translucent electro-optic termination of the single-mode fiber channel. Another important contribution aims to advance light- and radio-based sensing technologies towards networked versions that offer simultaneous communication and detecting capabilities.

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Tags:

Autonomous vehicles, Optical fiber communication, Telecommunications, Wireless communication, 6G, 6G mobile communication, Light emitting diodes, Low latency communication, Free-space optical communication, Optical communication terminals, Optical antennas, Optical fiber networks, Optical fiber sensors, Optical fibers, Wireless sensor networks

A Blockchain Framework for Incentivized Data Sharing in Autonomous Vehicle Networks*

Item Type Conference Paper
Author G. Olivieri
Author A. M. Mangini
Author M. Pia Fanti
Abstract Autonomous vehicles (AVs) continuously generate high-resolution sensor data on road conditions, infrastructure updates, and traffic dynamics. Despite their critical relevance for real-time navigation and urban planning, these datasets remain siloed within manufacturer-specific platforms. Motivated by the necessity to overcome such fragmentation, this paper introduces a novel decentralized, blockchain-based framework whose key innovation is a dynamic voting threshold integrated into a modular smart contract architecture. In our model, AVs can submit and validate road events –such as newly detected closures or construction sites– through a modular smart contract system employing dynamic voting thresholds that adapt acceptance criteria based on different factors. This allows urgent changes to achieve consensus while quickly minimizing malicious or erroneous reporting. Upon reaching a consensus regarding the specific event, the proposer is granted token-based incentives redeemable for operational cost reductions (e.g., charging or parking discounts). The proposed approach is validated via a Hardhat simulation on an Ethereum Virtual Machine compatible test network, demonstrating our design’s feasibility, robustness, and responsiveness under diverse scenarios.
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Extra Journal Abbreviation: 2025 11th International Conference on Control, Decision and Information Technologies (CoDIT)
Volume 1
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Tags:

Autonomous vehicles, Autonomous Vehicles, Blockchain, Urban planning, Traffic control, Decentralized Architecture, Distributed Validation, Token-based Incentives, Robustness, Roads, Vehicle dynamics, Real-time systems, Blockchains, Smart contracts, Virtual machines

A Comprehensive Survey on AI in Counter-Terrorism and Cybersecurity: Challenges and Ethical Dimensions

Item Type Journal Article

Author I. Syllaidopoulos

Author K. S. Ntalianis

Author I. Salmon

Abstract The advanced capabilities in threat detection and mitigation brought about by the rapid development of Artificial Intelligence (AI) significantly impact the fight against terrorism and cybersecurity threats. However, critical concerns such as algorithmic bias, data quality limitations, and governance challenges introduce significant obstacles to their deployment. This paper provides a comprehensive overview of AI methodologies, such as predictive analytics, Natural Language Processing (NLP), and machine learning architectures (e.g., Support Vector Machines – SVM and Long Short-Term Memory – LSTM), and optimization algorithms (e.g., Particle Swarm Optimization – PSO), assessing their effectiveness in security applications. Additionally, AI-based approaches to surveillance, misinformation management, and anomaly detection are explored, focusing on their impacts on national security. Beyond the technical aspects, the paper highlights ethical concerns and policy issues, putting forward frameworks such as Explainable Artificial Intelligence (XAI) and crowdsourced intelligence (Crosint) to ensure the responsible and transparent deployment of AI. The integration of the technical, ethical, and operational perspectives addressed in this research contributes to a holistic understanding of the potential of AI in cybersecurity, while also ensuring adherence to AI governance standards.

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Artificial intelligence, cybersecurity, Deep learning, machine learning, Machine learning, Natural language processing, Threat assessment, Surveys, Biological system modeling, Computer security, Ethics, counter-terrorism, explainable artificial intelligence (XAI), Fake news, misinformation management, natural language processing (NLP), threat detection

A Lightweight 5G-V2X Intra-Slice Intrusion Detection System Using Knowledge Distillation

Item Type Conference Paper

Author S. Hossain

Author A. Boualouache

Author B. Brik

Author S. -M. Senouci

Abstract As the automotive industry grows, modern vehicles will be connected to 5G networks, creating a new Vehicular-to-Everything (V2X) ecosystem. Network Slicing (NS) supports this 5G-V2X ecosystem by enabling network operators to flexibly provide dedicated logical networks addressing use case specific-requirements on top of a shared physical infrastructure. Despite its benefits, NS is highly vulnerable to privacy and security threats, which can put Connected and Automated Vehicles (CAVs) in dangerous situations. Deep Learning-based Intrusion Detection Systems (DL-based IDSs) have been proposed as the first defense line to detect and report these attacks. However, current DL-based IDSs are processing and memory-consuming, increasing security costs and jeopardizing 5G-V2X acceptance. To this end, this paper proposes a lightweight intrusion detection scheme for 5G-V2X sliced networks. Our scheme leverages DL and Knowledge Distillation (KD) for training in the cloud and offloading knowledge to slice-tailored lightweight DL models running on CAVs. Our results show that our scheme provides an optimal trade-off between detection accuracy and security overhead. Specifically, it can reduce security overhead in computation and memory complexity to more than 50% while keeping almost the same performance as heavy DL-based IDSs.

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Security, Deep learning, Privacy, 5G-V2X, IDS, Knowledge Distillation, Network slicing, Network Slicing, Biological system modeling, Industries, Knowledge engineering, Training

A Perspective Study Towards Biometric-based Rider Authentication Schemes For Driverless Taxis

Item Type Conference Paper

Author S. Gupta

Author B. Crispo

Abstract Driverless vehicles exploit Artificial Intelligence (AI) to offer rides to their users with minimal or no human input. Driverless vehicles as taxis are indeed evolving an entirely new transportation concept, Transportation-As-A-Service (TAAS), steered by well-proven client-server infrastructures. Typically, the servers, managed by multinational companies (e.g., Waymo), operate driverless taxis for public transportation, whereas the clients are simply the smartphone-based applications that can be used by customers for one-time registration and booking rides with driverless taxis anytime anywhere. In this paper, we perform a perspective study towards secure and usable biometric-based rider authentication schemes for driverless taxis. Our study is complemented by an online survey that comprises 75 responses from participants, worldwide. We published the survey on social media platforms like LinkedIn, Facebook, Twitter, and various public transport users groups to collect public poll for rider authentication for driverless taxis. Approximately, 90% of the overall participants in our survey either strongly agreed or agreed with the necessity to deploy biometric-based rider authentication for driverless taxis. Then, considering the survey's results, we proposed the design of a possible solution for riders biometric authentication.

Date 22-23 Sept. 2019

DOI 10.1109/3ICT.2019.8910310

Extra Journal Abbreviation: 2019 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)

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Conference Name 2019 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT)

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Tags:

Safety, Authentication, Public transportation, human-computer interaction, Sensors, Rider authentication, Face, physiological and behavioral biometric traits, Physiology, TAAS

A Provably Secure Lightweight Authentication Based on Elliptic Curve Signcryption for Vehicle-to-Vehicle Communication in VANETs

Item Type Journal Article

Author H. Shao

Author C. Piao

Abstract In vehicular ad hoc networks, the exchange of safety messages plays a crucial role in transportation efficiency and safety. However, safety messages are generally only signed rather than encrypted when broadcast, which lures malicious attackers to collect sensitive and private data. Obviously, an efficient privacy-preserving signcryption mechanism is essential for the timely and correct acceptance of safety messages. Currently, most state-of-the-art mechanisms satisfy security requirements, but performance is inefficient. Therefore, in this article, we design a provably secure lightweight authentication based on elliptic curve signcryption (EPSLA) for vehicle-to-vehicle communication, which can fulfill batch unsigncryption and a feasible security communication scheme in practical scenarios. The EPSLA scheme ensures security in relation to existential unforgeability against adaptive chosen message attacks and indistinguishability against adaptive chosen ciphertext attacks under the hardness assumption of the elliptic curve discrete logarithm problem in the random oracle model. Performance analysis shows that our scheme significantly performs better than the comparison schemes.

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Safety, Security, Authentication, Privacy, blockchain, certificateless signcryption, performance, Public key, Blockchains, Phase shift keying

A Review of Adversarial Machine Learning for Cyber security in Intelligent Systems using Image Classification & Signal Classification Analysis

Item Type Conference Paper

Author T. Jahan

Abstract Machine learning models with adversarial examples must be trained to differentiate between genuine and invading data. Cyber computing technology and tools face huge intrusion attacks and malicious programs. Tampered QR code for online payments, malicious face and biometric detection, incorrect facial recognition, object and sign detection are the adversarial examples in real-life scenarios. Convolutional neural networks often fail to image classification tasks due to high vulnerability. The Intelligent systems comprising computer vision, self-driving cars, radio signals, IoT Devices, industrial production and automation are prone to cyber attack with adversarial data in machine learning models. The goal of this paper is to summarize the various Convolutional Neural Network models and Recurrent Neural Network approaches used in machine learning for malicious image classification. The analysis aims to identify the area of attacks using adversarial machine learning in cyber physical systems and intrusion

detection systems.

Date 8-9 Dec. 2023

DOI 10.1109/ICTBIG59752.2023.10456232

Extra Journal Abbreviation: 2023 IEEE International Conference on ICT in Business Industry & Government (ICTBIG)

Pages 1-3

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Tags:

Convolutional neural networks, CNN, Adversarial machine learning, Face recognition, Reviews, Recurrent neural networks, Computational modeling, Cyber Security, Data models, Adversarial Machine Learning, Malware Detection, RNN

A Review Study of IEEE 802.11p On-Board Unit for V2X Deployment

Item Type Conference Paper

Author C. C. Fung

Author S. Yogarayan

Author S. F. Abdul Razak

Author A. Azman

Abstract Intelligent transportation systems (ITS) are increasing in popularity as a way to improve road safety and efficiency. A critical component of ITS is the use of wireless communication systems, such as the dedicated short-range communication (DSRC) IEEE 802.11p standard, to enable communication between vehicles and infrastructure. This project aims to evaluate the performance of an on-board unit (OBU) utilizing the IEEE 802.11p standard in a vehicular communication system. The project contributes to a better understanding of the capabilities and limitations of the IEEE 802.11p standard in the context of ITS applications. The project will certainly also be useful for researchers and the community working in this field. Besides, the project also provides valuable insights into the performance of an OBU utilizing the IEEE 802.11p standard and can be used to convey the optimization and improvement of vehicular communication systems for the use of ITS applications.

Date 23-24 Aug. 2023

DOI 10.1109/ICoICT58202.2023.10262808

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Pages 165-171

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Tags:

Security, Road safety, V2X, Privacy, Transportation, Pedestrians, communication, Protocols, IEEE 802.11p, IEEE 802.11p Standard, OBU

An empirical study on automotive cyber attacks

Item Type Conference Paper

Author A. Singh

Author M. Singh

Abstract There has been various technological advancements in the field of autonomous vehicles. Which increase the possibility of the introduction of these vehicles in the market in some years. Moreover, then we will surround all with numerous autonomous vehicles and other intelligent systems making our life easier. They will include various networked embedded systems connected to internet running on numerous algorithms. These systems may contain vulnerabilities and threats that can be exploited. Therefore, these systems must be developed in order to minimize all the possible threats and vulnerabilities. In this study, we will majorly focus on the Cyber-Security vulnerabilities and threats that are possible in automotive electronics.

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DOI 10.1109/WF-IoT.2018.8355124

Extra Journal Abbreviation: 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

Pages 47-50

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Conference Name 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

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Tags:

Security, Automobiles, Cyber-attacks, Cyber-Physical Systems, Servers, Companies, Automotive, Automotive electronics, Automotive Electronics, Threats, Vulnerabilities

An IoT-based Two-Level Authenticated Driving System

Item Type Conference Paper

Author K. Kavitha

Author J. G

Author T. P. Ranga Vital

Author C. C. Sekhar

Author R. Vemala

Author S. Marpu

Abstract In recent years, there has been significant growth in complex electronic computer control systems for vehicles, particularly in the development of smart cars and autonomous vehicles. The term internet of things (IoT) refers to the network of "things" in order to connect and exchange data through sensors and softwares without the need for human interaction. This paper targets the critical issue of minors and unlicensed drivers driving vehicles without permission, in order to reduce possible accidents and encourage responsible ownership of vehicles. Driving without a license endangers everyone's safety since these drivers are ignorant of traffic regulations, safe driving practices, and the fact that they won't have insurance. Our solution uses Radio Frequency Identification (RFID) technology and biometric fingerprint identification with cloud integration to develop a strict vehicle access control system. These technologies are integrated with an Arduino microcontroller. We have evaluated our system using different metrics like user permission simulation and security alerts. This system provides a safe and efficient way of managing vehicle access, significantly minimizing the risk, with the main aim of permitting only authorized and responsible drivers to drive the bike. This successfully stops minors from using unlawful collisions caused by unlicensed and unskilled drivers.

Date 18-19 July 2025

DOI 10.1109/CIACON65473.2025.11189487

Extra Journal Abbreviation: 2025 International Conference on Computing, Intelligence, and Application (CIACON)

Pages 1-7

Proceedings Title 2025 International Conference on Computing, Intelligence, and Application (CIACON)

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Tags:

Safety, Accidents, Vehicles, Road safety, IoT, Internet of Things, vehicle, Microcontrollers, Software, Access control, Arduino uno, Authorized Driving System, biometric, Fingerprint recognition, Radiofrequency identification, RFID

Applied Post-Quantum Secure Method for IoT Devices: A Case Study for Autonomous Vehicles Communication

Item Type Conference Paper

Author G. R. Figlarz

Author F. P. Hessel

Abstract Autonomous vehicles (AV) have as one of their characteristics the constant exchange of data with other vehicles, radio base stations, pedestrians and infrastructure. Despite this data exchange using the most modern security techniques, vehicles have been the target of constant cyber-attacks. Emerging technologies such as quantum computing have been tested to minimize these attacks. Particularly, this work investigates the possibility of using post-quantum computing techniques in IoT (Internet of Things) devices, more specifically in AVs. Considering the functionalities of a vehicle nowadays, it is possible to say that a vehicle is composed of a set of IoT devices. Therefore, in this paper, we propose a post-quantum secure method to be used in IoT devices for data exchange between AVs and a 5G base station. In this method, the post-quantum algorithm NTRU is implemented in an edge computing IoT architecture between the source of the CVs data and the edge layer, which is a vulnerable spot for quantum-attacks. We simulated the implementation of the algorithm in a case study and monitored how the algorithm performed. The proposed methodology was capable of providing secure communication between IoT and edge layer. However, the message length and parameters from the algorithm can slow down the communication.

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Extra Journal Abbreviation: 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)

Pages 1-6

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Tags:

autonomous vehicles, Autonomous vehicles, Computer architecture, IoT, Internet of Things, Cryptography, edge computing, Quantum computing, Performance evaluation, Base stations, NTRU, post-quantum cryptography

Approach of an advanced autonomous vehicle with data optimization and cybersecurity for enhancing vehicle's capabilities and functionality for smart cities

Item Type Conference Paper

Author V. Saini1

Author A. Jain

Author A. Dodia

Author M. K. Prasad

Abstract Autonomous competition is a brand-new area of technological advancement that has just emerged because of the increasing acceptance of self-driving vehicles, particularly after 2015. For high-performance automobiles designed to drive autonomously in extremely unpredictable, dynamic, and hostile contexts,

researchers are creating hardware and software algorithms that work precisely. Our Research mainly focused on the balanced approach for System Architecture, Data optimization in autonomous vehicle to enhance the smart cities. The most important abilities and features such as communication, autonomous path planning, path following control, navigation, obstacle detection and depth estimation are regarded as the foundational knowledge of agents. These conditions include high speeds, high safety and security, data optimization, slow reaction times, and antagonistic real-world surroundings. This study is the first comprehensive and deep analysis of the research on autonomous vehicles, including their architectural design, data optimization, safety & security and Internet of Things for high end applications. Autonomous vehicle cybersecurity is also important to assure the safe and secure deployment of these vehicles. Industry 4.0 can assist maintain the safety and security of autonomous vehicles by deploying effective cybersecurity solutions that address the essential qualities of confidentiality, integrity, availability, and non-repudiation. Finally, this research study emphasizes the balanced integration approach of System Architecture, Safety & security and the "software-hardware" co-evolution to the current level while providing a thorough review of the current autonomous vehicle systems.

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Extra Journal Abbreviation: 7th IET Smart Cities Symposium (SCS 2023)

Volume 2023

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Ascertaining Factors Affecting Autonomous Driving in New Zealand - A Framework for HMI Design

Item Type Conference Paper

Author A. U. Rehman

Author A. Ghaffarianhoseini

Author N. Naismith

Author J. Tookey

Author A. Ghaffarianhoseini

Author S. Urréhman

Abstract Autonomous Vehicles [AVs] will emerge as a powerful catalyst, thus forming a potentially disruptive technology opening doors to technological, socio-cultural and legal challenges. Present-day research is mainly focused on technological orientation, lacking the predominant social and behavioural linkages. NZ being a testbed of innovative technologies, is facing significant challenges in growing population and traffic congestions. Auckland ranks 47th globally in terms of traffic congestion costing \$1.3b a year to the economy. Therefore, harnessing AV technology will be a significant step to remain competitive. The people's readiness for acceptance influences AVs successful implementation. In this background, it is quintessential to ascertain and evaluate critical factors affecting

the successful deployment of AVs in NZ. This exploratory research study is a part of a larger project of realizing a trust dynamics and governance framework for humanizing driverless technology. It is being carried out in collaboration with BMW NZ Group using Autonomous Level 2 Vehicle in Auckland on high, medium and low-density roads with level 3 functions mostly. Drawing on systematic qualitative evidence, this study attempted to find out key determinants/variables affecting user acceptance, including anthropomorphism, training, feedback, safety, security, privacy, customization and adaptive automation for AVs deployment in NZ and their role in garnering users' trust besides suggesting an HMI Autonomous Driving Events Relationship Framework for AVs.

Date 18-20 Dec. 2021

DOI 10.1109/ICVISIP54630.2021.00011

Extra Journal Abbreviation: 2021 5th International Conference on Vision, Image and Signal Processing (ICVISIP)

Pages 1-8

Proceedings Title 2021 5th International Conference on Vision, Image and Signal Processing (ICVISIP)

Conference Name 2021 5th International Conference on Vision, Image and Signal Processing (ICVISIP)

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Tags:

Safety, Security, Vehicle dynamics, Sociology, Systematics, Training, Autonomous Driving (AD), Autonomous Vehicle (AV), Human-Machine Interaction (HMI), Signal processing

Automotive Knowledge Base for Supporting Vehicle-SOC Analysts

Item Type Conference Paper

Author V. S. Barletta

Author D. Caivano

Author M. De Vincentiis

Author A. Pal

Author F. Volpe

Abstract With the introduction of more and more ECUs and components in vehicles, the automotive industry is moving towards increasingly connected and autonomous driving. This means not only more functionality available but also an increased risk to vehicle security and driver safety. Therefore, it is necessary to improve the existing solutions and create new techniques and methods not only to detect cyber attacks but also to mitigate and respond with appropriate actions. The paper proposes a Knowledge Base in Automotive to be integrated within a Vehicle Security Operation Center (V-SOC) in order to provide support to analysts in being able to reconstruct the attack kill chain and understand the impact of the attack on other components. The idea is to exploit, at this stage of the research, existing taxonomies in the literature on automotive attacks in order to obtain the classification of existing attacks, the automotive security development process, and the decomposition of incidents that consist of multiple attack steps.

Date 25-27 Oct. 2023

DOI 10.1109/MetroXRINE58569.2023.10405622

Extra Journal Abbreviation: 2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRINE)

Pages 960-965

Proceedings Title 2023 IEEE International Conference on Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRINE)

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Tags:

Safety, Security, cybersecurity, Vehicles, Automotive engineering, Knowledge based systems, automotive, Taxonomy, knowledge base, Neural engineering

Autonomous and Security-Aware Dynamic Vehicular Platoon Formation

Item Type Conference Paper

Author D. Phillips

Author T. Halabi

Author M. Zulkernine

Abstract As vehicles increasingly embed digital systems, new security vulnerabilities are also being introduced. Computational constraints make it challenging to add security oversight layers on top of core vehicle systems, especially when the security layers rely on additional deep learning models for anomaly detection. To improve security-aware decision-making for autonomous vehicles (AV), this paper proposes a bi-level security framework. The first security level consists of a one-shot resource allocation game that enables a single vehicle to fend off an attacker by optimizing the configuration of its intrusion prevention system based on risk estimation. The second level relies on a reinforcement learning (RL) environment where an agent is responsible for forming and managing a platoon of vehicles on the fly while also dealing with a potential attacker. We solve the first problem using a minimax algorithm to identify optimal strategies for each player. Then, we train RL agents and analyze their performance in forming security-aware platoons. The trained agents demonstrate superior performance compared to our baseline strategies that do not consider security risk.

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Extra Journal Abbreviation: 2023 10th International Conference on Dependable Systems and Their Applications (DSA)

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Tags:

Deep learning, Decision making, security, reinforcement learning, Reinforcement learning, Heuristic algorithms, game theory, platoon formation, risk estimation, Vehicular platoons, Digital systems, Estimation, Games

Autonomous Driving Security: A Comprehensive Threat Model of Attacks and Mitigation Strategies

Item Type Conference Paper

Author M. A. Hoque

Author R. Hasan

Abstract Autonomous vehicles (AVs) are envisioned to enhance safety and efficiency on the road, increase productivity, and positively impact the urban transportation system. Due to recent developments in autonomous driving (AD) technology, AVs have started moving on the road. However, this promising technology has many unique security challenges that have the potential to cause traffic accidents. Though some researchers have exploited and addressed specific security issues in AD, there is a lack of a systematic approach to designing security solutions using a comprehensive threat model. A threat model analyzes and identifies potential threats and vulnerabilities. It also identifies the attacker model and proposes mitigation strategies based on known security solutions. As an emerging cyber-physical system, the AD system requires a well-designed threat model to understand the security threats and design solutions. This paper explores security issues in the AD system and analyzes the threat model using the STRIDE threat modeling process. We posit that our threat model-based analysis will help improve AVs' security and guide researchers toward developing secure AVs.

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Tags:

autonomous vehicles, Safety, Transportation, security, Cyber-physical systems, Threat modeling, Roads, Systematics, Productivity, threat model

Autonomous vehicle ultrasonic sensor vulnerability and impact assessment

Item Type Conference Paper

Author B. S. Lim

Author S. L. Keoh

Author V. L. L. Thing

Abstract Vehicles today are relying more on technologies to bring about fully autonomous features. The conventional wirings within are being simplified into a network of electronic components, and this network is controlled via advanced sensing of the environment to make decisions in real-time. However, with the heavy reliance on the sensor readings, any inaccurate reading from the sensors could result in decisions that may cause life-threatening incidents. As such, this research focuses on the in-depth assessment of potential vulnerabilities of an important and commonly used obstacle sensing device, which is the ultrasonic sensor, in modern as well as autonomous vehicles. This research will help bring awareness to the car manufacturers and AV researchers so as to mitigate such issues.

Date 5-8 Feb. 2018

DOI 10.1109/WF-IoT.2018.8355132

Extra Journal Abbreviation: 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

Pages 231-236

Proceedings Title 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

Conference Name 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

Autonomous vehicles, Sensors, Roads, Acoustics, Light emitting diodes, Pins

Bridging the Gaps: A Comparative Analysis of ISO 21434, ISO 26262 and Machine Learning in Autonomous Vehicles

Item Type Conference Paper

Author S. Daiene de Oliveira Bastos

Author K. Castelo Branco

Author A. Luiz de Oliveira

Abstract The integration of Machine Learning (ML) in Autonomous Vehicles (AV s) raises challenges not fully addressed by current automotive standards, specifically ISO 26262 (functional safety) and ISO/SAE 21434 (cybersecurity). This paper reviews gaps in these standards concerning ML-enabled systems, focusing on explainability, data lifecycle, runtime assurance, and ethical compliance. To bridge these gaps, we propose the ML-Aware Compliance Bridge, a four-layer framework covering: (i) Data Lifecycle Management, (ii) Model Governance, (iii) Runtime Assurance, and (iv) Ethics and Explainability. The framework maps standard deficiencies to actionable recommendations, supporting the development of safer and more trustworthy AI -driven autonomous systems.

Date 13-15 Oct. 2025
DOI 10.1109/SBR/WRE66973.2025.11249566
Extra Journal Abbreviation: 2025 Brazilian Symposium on Robotics (SBR) and 2025 Workshop on Robotics in Education (WRE)
ISBN 2643-685X
Pages 25-30
Proceedings Title 2025 Brazilian Symposium on Robotics (SBR) and 2025 Workshop on Robotics in Education (WRE)
Conference Name 2025 Brazilian Symposium on Robotics (SBR) and 2025 Workshop on Robotics in Education (WRE)
Date Added 20/09/2026, 7:46:50 pm
Modified 20/09/2026, 7:46:50 pm

Tags:

Autonomous vehicles, Safety, Machine learning, ISO Standards, Robots, Machine Learning, Reviews, ISO/SAE 21434, ISO 26262, Ethics, Standards, Bridges, Runtime

Can social media shape the security of next-generation connected vehicles?

Item Type Conference Paper
Author N. Scarano
Author L. Mannella
Author A. Savino
Author S. D. Carlo
Author P. D. Torino
Abstract abstract-The increasing adoption of connectivity and electronic components in vehicles makes these systems valuable targets for attackers. While automotive vendors prioritize safety, there remains a critical need for comprehensive assessment and analysis of cyber risks. In this context, this paper proposes a Social Media Automotive Threat Intelligence (SOCMATI) framework, specifically designed for the emerging field of automotive cybersecurity. The framework leverages advanced intelligence techniques and machine learning models to extract valuable insights from social media. Four use cases illustrate The framework's potential by demonstrating how it can significantly enhance threat assessment procedures within the automotive industry.
Date 3-5 July 2024
DOI 10.1109/IOLTS60994.2024.10616053
Extra Journal Abbreviation: 2024 IEEE 30th International Symposium on On-Line Testing and Robust System Design (IOLTS)
ISBN 1942-9401
Pages 1-4
Proceedings Title 2024 IEEE 30th International Symposium on On-Line Testing and Robust System Design (IOLTS)
Conference Name 2024 IEEE 30th International Symposium on On-Line Testing and Robust System Design (IOLTS)
Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

Safety, Machine learning, Social networking (online), Automotive Security, Cyber Threat Intelligence, Online Social Networks, Threat assessment, Shape, Industries, System analysis and design

CarLINK: AI Driven Traffic Adviser for Automobiles

Item Type Conference Paper

Author J. Gangurde

Author G. Ubale

Author P. Ghante

Author A. Fulsundar

Author S. Divate

Author V. Dalvi

Abstract The transformation in autonomous vehicles has seen huge advancements in innovation, with tech organizations heavily investing in research and development in self-driving technology. The earlier ten years saw an important ascent in the progression of autonomous vehicles, fundamentally supported by improvements in deep learning and artificial intelligence. However, when we saw the boundless reception of autonomous vehicles on our streets, we encountered many anomalies in the framework and some accident cases as well. Decision-making is a critical and complex aspect of self-driving cars, being essential to guaranteeing the effectiveness, safety, and adaptability of autonomous cars. Effective decision-making is at the core of the success and acceptance of self-driving cars. The development of robust algorithms, ethical considerations, adaptability to various conditions, and compliance with legal standards are all crucial elements that contribute to the overall functionality and safety of autonomous vehicles. The crust of any self-driving vehicle is its neural network as it is ultimately responsible for decision-making. As the traditional Neural Network Architectures fail to achieve efficient decision-making in self-driving cars we propose a new Neural Network to achieve effective decision making thus lowering the number of accidents gradually leading to safe driving.

Date 7-8 June 2024

DOI 10.1109/ICICET59348.2024.10616337

Extra Journal Abbreviation: 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

Pages 1-7

Proceedings Title 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

Conference Name 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

autonomous vehicles, Safety, Transportation, Decision making, Traffic control, Technological innovation, Neural networks, traffic congestion, decision making, Autonomous automobiles, model, neural network

CDBV: A Driving Dataset With Chinese Characteristics From a Bike View

Item Type Journal Article

Author Y. He

Author H. -B. Yang

Author S. -J. Wang

Abstract Public datasets are valuable for the development of the learning-based algorithms. In the field of autonomous driving, almost all the existing public datasets are collected by manually driving cars, and most of them are contributed by American and European researchers and represent the traffic scenarios typical for these countries. To diversify the public driving datasets and to provide data support for researches on the self-driving vehicles moving on bike lanes, such as wheelchairs, bikes, and some robots, we present the Chinese Driving from a Bike View (CDBV), which is a driving dataset collected by a bike in Beijing, China. The CDBV dataset contains annotated images, raw videos, and code tools. The annotations are for the object detection tasks and especially focus on the categories with Chinese characteristics, including the express tricycles, the motorbikes specializing in takeouts, the bicycle landmarks, the zebra crossings, and the traffic lights. We totally annotated 40 064 traffic elements in 13 427 images. The CDBV dataset is publicly available at <http://www.sujingwang.name/CDBV.html>.

Date 2019

DOI 10.1109/ACCESS.2019.2911562

Volume 7

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Journal Abbr IEEE Access

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Tags:

Autonomous vehicles, Automobiles, Task analysis, China, Object detection, autonomous driving, Urban areas, Videos, collected by a bike, Driving dataset, Motorcycles, object detection, self-driving wheelchair

Challenges in Cyber Physical Social Systems and Internet of Things

Item Type Conference Paper

Author S. Revathi

Author J. Raja

Author M. Mohanraj

Author K. Malathi

Author B. Mallala

Author R. G. Vidhya

Abstract This research analyses the complex dynamics of Cyber-Physical-Social Systems (CPSS), encompassing cyber-physical systems, cybersecurity, the Internet of Things (IoT), and social media. By exploring the interactions among these elements, this study analyses the vulnerabilities, risks, and potential benefits within the CPSS framework. This research study analyses the evolving landscape of cybersecurity by considering various aspects such as data integrity, privacy in the IoT era, and the influence of social media on user behaviour and influence dynamics. The study also introduces innovative strategies to enhance CPSS resilience, including methods to mitigate cyber threats and enhance overall system integrity. Additionally, we investigate how social media shapes CPSS behaviours, offering valuable insights into influence dynamics. This research advances the comprehensive understanding of CPSS to support future studies and advancements in the protection and improvement of interconnected systems.

Date 18-20 Sept. 2024

DOI 10.1109/ICOSEC61587.2024.10722103

Extra Journal Abbreviation: 2024 5th International Conference on Smart Electronics and Communication (ICOSEC)

Pages 409-413

Proceedings Title 2024 5th International Conference on Smart Electronics and Communication (ICOSEC)

Conference Name 2024 5th International Conference on Smart Electronics and Communication (ICOSEC)

Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

Data privacy, Cybersecurity, Social networking (online), Internet of Things, Predictive analytics, Social Media, Internet of Things (IoT), Shape, Media, Real-time systems, Resilience, Protection, Cyber-Physical Systems, Monitoring, Cyber-Physical-Social Systems (CPSS)

Cloud-SPHERE: A Security Approach for Connected Unmanned Aerial Vehicles

Item Type Conference Paper

Author M. Rodrigues

Author D. F. Pigatto

Author K. R. L. J. C. Branco

Abstract Unmanned Aerial Vehicles (UAVs) have gained popularity and are current applied in many fields. They have benefited from the recent technology development and have now increased connectivity, which allows them to achieve higher levels of mission complexity and coverage. Among others, security is

considered one of the greatest challenges for UAV usage acceptance from the general public, and it scales when this vehicle can communicate with other vehicles and/or systems. This paper presents Cloud-SPHERE, a security approach for unmanned vehicles to communicate and provide services securely, targeting their integration into the IoT ecosystem. The design and operation is outlined and preliminary prototyping results are presented.

Date 12-15 June 2018

DOI 10.1109/ICUAS.2018.8453302

Extra Journal Abbreviation: 2018 International Conference on Unmanned Aircraft Systems (ICUAS)

ISBN 2575-7296

Pages 769-778

Proceedings Title 2018 International Conference on Unmanned Aircraft Systems (ICUAS)

Conference Name 2018 International Conference on Unmanned Aircraft Systems (ICUAS)

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Modified 20/09/2026, 7:46:50 pm

Tags:

Safety, Security, Task analysis, Computer architecture, Cloud computing, Unmanned aerial vehicles

Combined Cloud and Deep Learning Models for Enhanced Decision Support

Item Type Conference Paper

Author V. A

Author M. R. Borade

Author S. Kannimuthu

Author A. Singla

Author X. M. Raajini

Author S. Rampal

Abstract Cloud computing and deep learning are changing decision-support systems. This combination was driven by cloud computing and deep learning models' complimentary data processing capabilities. Traditional decision-support systems typically fail due to the complexity and variety of current information, particularly unstructured and semi-structured data. Innovative methods are required to overcome this restriction and exploit big data's immense potential. Cloud computing is essential for handling and storing huge amounts of data because to its design, scalability, and storage capacity. Company data is securely accessed, processed, and stored while being cost-effective and flexible. Deep learning models—a subset of AI—offer a great toolbox of neural networks and algorithms for analyzing complex data patterns. These models are versatile, train well, and can find patterns in unstructured data. However, ideal outcomes demand a lot of processing power. Deep learning and cloud computing operate together better than traditional decision-support systems. Instead of static, rule-based models, the decision-support system may now react to complicated, ever-changing data. Modern cloud-based algorithms enable organizations to make accurate, personalized choices in real time. These changes will affect healthcare, finance, manufacturing, and autonomous automobiles. Convergence causes actual concerns. Data security, privacy, and equality are key concerns. Cloud systems

are difficult to balance between usability and safety. Since deep learning algorithms may be prejudiced and provide unfair outcomes, we must continue to promote fairness and transparency. Additional challenges to the effective implementation of these converged systems include controlling scalability, ensuring compliance with laws, and addressing interoperability.

Date 13-14 Dec. 2024

DOI 10.1109/ICTBIG64922.2024.10911261

Extra Journal Abbreviation: 2024 IEEE 4th International Conference on ICT in Business Industry & Government (ICTBIG)

Pages 1-6

Proceedings Title 2024 IEEE 4th International Conference on ICT in Business Industry & Government (ICTBIG)

Conference Name 2024 IEEE 4th International Conference on ICT in Business Industry & Government (ICTBIG)

Date Added 20/09/2026, 7:46:49 pm

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Tags:

Data privacy, Artificial intelligence, Decision support systems, Big data, Deep learning, Machine learning, Data security, Cloud computing, Real-time systems, Scalability, Computational modeling, Government, Data models, Data processing, Data analysis, Decision support

Comparative Study of Attention among Drivers with Varying Driving Experience

Item Type Conference Paper

Author B. Adhikari

Author Z. Durić

Author D. Wijsekera

Author B. Yu

Abstract Advancements in Intelligent Transportation Systems (ITS) enhance driver safety, comfort, and traffic flow. However, because human drivers remain integral to operating human-driven vehicles, their perception and attention are of significant importance in real-world traffic conditions. Previous studies have highlighted the increased accident susceptibility of novice drivers, but there has been limited research on how experience and perception affect each other. This study analyzes driver behavior in real-world traffic conditions based on driver experience, time of day, traffic congestion, and route familiarity. Our findings indicate that novice drivers exhibit lower spatial awareness, while intermediate drivers demonstrate improved awareness in moderate traffic. Conversely, experienced drivers consistently perform well across all conditions. Familiarity with routes enhances performance for all drivers, with experienced drivers adapting more quickly in unfamiliar situations. The study also contributes to existing data by providing continuous driver fixations in various driving scenarios and gaze-enhanced saliency maps, gaze-focused object detection, and saliency ranking to advance driver perception and autonomous vehicles.

Date 2-5 June 2024

DOI 10.1109/IV55156.2024.10588683
Extra Journal Abbreviation: 2024 IEEE Intelligent Vehicles Symposium (IV)
ISBN 2642-7214
Pages 1353-1360
Proceedings Title 2024 IEEE Intelligent Vehicles Symposium (IV)
Conference Name 2024 IEEE Intelligent Vehicles Symposium (IV)
Date Added 20/09/2026, 7:46:50 pm
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Tags:

Autonomous vehicles, Safety, Accidents, Vehicles, Traffic congestion, Object detection, Advance Driver Assistance Systems(ADAS), and driver perception, Driver Behavior Analysis (DBA), saliency

Cooperative Intelligent Transport Systems Solution: a Cooperative Safety Application

Item Type Conference Paper

Author N. Dimokas

Author P. Spanidis

Author I. Tsetsinas

Author I. Gkragkopoulos

Author M. Gemou

Author P. Karvelis

Abstract Recent advances in wireless communications enable the communication between vehicles, infrastructure and road users. Although the vehicles become safer and smarter due to Advanced Driver Assistance Systems, Cooperative Intelligent Transport Systems have an important role to ensure that every vehicle recognizes properly the road environment. Cooperative Intelligent Transport Systems applications aims to collect data from different resources like sensors, cameras, vehicles, road infrastructure, road users and analyze them to improve traffic management and road safety. The proposed system provides an innovative technological solution that implements Cooperative Intelligent Transport Systems applications without substantial and costly roadway interventions, including novel Vehicle to Infrastructure, Vehicle to Vehicle and Variable Message Signs functions. This is achieved via low-cost, integrated strips on the road pavement coupled with a cooperative communication framework. The paper presents the overall system and the Cooperative Safety application. The results attested the realization of the system with low latency.

Date 23-25 Sept. 2022

DOI 10.1109/SEEDA-CECNSM57760.2022.9932930

Extra Journal Abbreviation: 2022 7th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)

Pages 1-8

Proceedings Title 2022 7th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)

Conference Name 2022 7th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)

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Tags:

Safety, Road safety, Privacy, Social networking (online), cooperative awareness, cooperative intelligent transport system, Low latency communication, Sensor systems and applications, sensors, strip, Strips, vehicle to infrastructure, vehicle to vehicle, wireless communication

Cyber physical surveillance system for Internet of Vehicles

Item Type Conference Paper

Author D. Singh

Author G. Tripathi

Author S. C. Shah

Author R. da Rosa Righi

Abstract Internet of Vehicle (IoV) is an essential part of the Intelligent Transportation system (ITS) which is growing exponentially in the automotive industry domain. The term IoV is used in this paper for Internet of Vehicles. IoV is conceptualized for sharing traffic, safety and several other vehicle-related information between vehicles and end user. In recent years, the number of connected vehicles has increased all over the world. Having information sharing and connectivity as its advantage, IoV also faces the challenging task in the cybersecurity-related matters. The future consists of crowded places in an interconnected world through wearable's, sensors, smart phones etc. We are converging towards IoV technology and interactions with crowded space of connected peoples. However, this convergence demands high-security mechanism from the connected crowd as well as other connected vehicles to safeguard of proposed IoV system. In this paper, we coin the term of smart people crowd (SPC) and the smart vehicular crowd (SVC) for the Internet of Vehicles (IoV). These specific crowds of SPC and SVC are the potential cyber attackers of the smart IoV. People connected to the internet in the crowded place are known as a smart crowd. They have interfacing devices with sensors and the environment. A smart crowd would also consist of the random number of smart vehicles. With the future converging in to the smart connected framework for crowds, vehicles and connected vehicles, we present a novel cyber-physical surveillance system (CPSS) framework to tackle the security threats in the crowded environment for the smart automotive industry and provide the cyber security mechanism in the crowded places. We also describe an overview of use cases and their security challenges on the Internet of Vehicles.

Date 5-8 Feb. 2018

DOI 10.1109/WF-IoT.2018.8355218

Extra Journal Abbreviation: 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

Pages 546-551

Proceedings Title 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)

Conference Name 2018 IEEE 4th World Forum on Internet of Things (WF-IoT)
Date Added 20/09/2026, 7:46:50 pm
Modified 20/09/2026, 7:46:50 pm

Tags:

Automotive engineering, Entropy, Cloud computing, Surveillance, Intelligent sensors

Deep Learning for Lane Discipline: AI-Powered Traffic Enforcement in Smart Cities

Item Type Conference Paper
Author T. Vaishnavi
Author P. Paulraj
Author K. S. Krishnamohan
Author K. T. Al-Assaf
Author A. Dutt
Author G. Karthikeyan
Abstract Connected vehicles are poised to revolutionize future transportation systems by enabling seamless communication and real-time data exchange, enhancing safety, and optimizing traffic flow. This study proposes a novel data-driven AI framework for real-time lane enforcement in connected vehicle ecosystems, aiming to ensure compliance with lane restrictions and traffic rules while minimizing manual intervention. The proposed algorithm integrates advanced AI techniques, stochastic modelling, and real-time monitoring to create a robust and scalable model. By combining probabilistic reasoning, adaptive learning, and optimization strategies, the framework achieves higher efficiency and accuracy than existing methods. The experimental results demonstrate the superior performance of the proposed methodology in reducing violations, improving response times, and enhancing overall traffic safety, making it a promising solution for modern intelligent transportation systems. Keywords: Connected vehicles, AI-driven framework, real-time lane enforcement, stochastic modelling, traffic safety, probabilistic reasoning, adaptive learning, intelligent transportation systems.
Date 7-8 Aug. 2025
DOI 10.1109/ICCPCT65132.2025.11176748
Extra Journal Abbreviation: 2025 8th International Conference on Circuit, Power & Computing Technologies (ICCPCT)
Pages 530-536
Proceedings Title 2025 8th International Conference on Circuit, Power & Computing Technologies (ICCPCT)
Conference Name 2025 8th International Conference on Circuit, Power & Computing Technologies (ICCPCT)
Date Added 20/09/2026, 7:46:50 pm
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Tags:

Safety, Artificial intelligence, traffic safety, Connected vehicles, Biological system modeling, Real-time systems, Cognition, Adaptation models, adaptive learning, Adaptive learning, AI-driven framework, Infrastructure management, Intelligent Transportation System (ITS), Probabilistic logic, probabilistic reasoning, real-time lane enforcement, stochastic modelling, Stochastic processes

Developing a Smart Pedestrian Network Big Data Platform for Municipal Organizations

Item Type Conference Paper

Author G. Papageorgiou

Author C. Petrakis

Author A. Maimaris

Abstract In our Digital Economy, great opportunities but also challenges are emerging for today's organizations. To successfully implement information system solutions involving a variety of data sources, many aspects should be considered. These revolve around the choice of platform strategy, which should handle aspects such as the provision of Application Programming Interfaces (API), privacy and security concerns and how open access to data should be. This paper examines design issues for developing a digital ecosystem for municipal organizations by considering the opportunities presented by Big Data. The main focus is to suggest solutions for the deployment of an envisioned Smart Pedestrian Network (SPN) system. The proposed solution combines multiple subsystems, such as Geographical Information Systems, intelligent agents, sensor systems, crowdsourcing, and social networks to create an integrated big data approach for the urban environment. Such an ecosystem is exceptionally useful for the effective involvement of municipality stakeholders, such as members of the public community, management, city planning, public works maintenance teams as well as local businesses.

Date 20-22 Dec. 2018

DOI 10.1109/EECS.2018.00058

Extra Journal Abbreviation: 2018 2nd European Conference on Electrical Engineering and Computer Science (EECS)

Pages 274-278

Proceedings Title 2018 2nd European Conference on Electrical Engineering and Computer Science (EECS)

Conference Name 2018 2nd European Conference on Electrical Engineering and Computer Science (EECS)

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Tags:

Data privacy, Ecosystems, Crowdsourcing, Social networking (online), Urban areas, Big Data, Smart Cities, Sensors, Organizations, Intelligent Systems, Municipal Organizations, Platform Strategies

Development of AI-Based Microdrone Capabilities for Multifunctional Aerial Applications

Item Type Conference Paper

Author V. -D. Popescu

Author M. Giurgiu

Author M. Cserei

Abstract This paper presents the design, implementation, and experimental evaluation of a new concept for a multi-drone system capable of executing complex aerial missions in both indoor and outdoor environments. The novelty of the proposed architecture consists in integrating ultra-wideband communication, onboard Artificial Intelligence (AI) for object detection, and a web-based control interface to coordinate multiple microdrones of the type Crazyflie 2.1. Key functionalities include real-time video surveillance, autonomous navigation to predefined coordinates, precise indoor positioning using Time Difference of Arrival (TDoA), optical flow-based stability, and onboard object detection. Experimental results demonstrate the system's robustness, adaptability, and potential for real-world applications such as security monitoring, industrial inspection, and intelligent autonomous operations in GPS-denied environments.

Date 25-26 Nov. 2025

DOI 10.1109/TELFOR67910.2025.11314275

Extra Journal Abbreviation: 2025 33rd Telecommunications Forum (TELFOR)

ISBN 2994-5828

Pages 1-4

Proceedings Title 2025 33rd Telecommunications Forum (TELFOR)

Conference Name 2025 33rd Telecommunications Forum (TELFOR)

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Tags:

Security, Time difference of arrival, Video surveillance, Real-time systems, Telecommunications, Software, Solids, Smart agriculture, Edge AI, Inspection, Microdrone, Ultra wideband communication, UWB Communication

Digital Disruption at the Edge: A Review of Barriers to Drone Technology in Last-Mile Delivery

Item Type Conference Paper

Author M. S. Dildar Awan

Author S. Menon

Author S. R. Shah

Abstract Emerging technologies in the era of digital supply chain have fostered the development of innovative strategies to enhance the last mile delivery system within logistics to improve overall efficiency within the supply chain. Among these, drone technology is gaining prominence due to its distinctive attributes

such as rapid delivery capabilities, cost effectiveness and high operational flexibility thus providing competitive edge to the businesses to respond to customer delivery requirements. While drone technology contributes to achieving a responsive supply chain, several challenges impact their commercial deployment and scalability of drones logistics network. This paper identifies the critical challenges that impact the implementation and utilisation of the drone technology within last mile delivery encompassing various external and internal factors. The findings underscore the necessity for a nuanced understanding of these multifaceted barriers to ensure successful integration of drone into the last mile delivery system for an efficient and responsive supply chain.

Date 20-22 Oct. 2025

DOI 10.1109/ICTMOD66732.2025.11372024

Extra Journal Abbreviation: 2025 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)

ISBN 2159-5119

Pages 1-6

Proceedings Title 2025 IEEE International Conference on Technology Management, Operations and Decisions (ICTMOD)

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Tags:

Safety, Drones, Drone technology, Technological innovation, Costs, Logistics, Supply chains, Urban areas, Reviews, Scalability, Technology management, Digital Supply Chain, Last Mile Delivery, Weather

Digital Health for the post-COVID-19 Pandemic in India: Emerging Technologies for Healthcare

Item Type Conference Paper

Author A. Ramachandran

Author S. N. Sarbadhikari

Abstract COVID-19 has led to unprecedented utilization of digital solutions to combat the pandemic. The purpose of the current article is to present an overview of digital technology adoption to handle COVID in the country and to explore the challenges in its continuous adoption in the current post-COVID era. This work is based on the articles, news and reports collected from literature databases like Pubmed, Ebsco, news reports, and blog and government websites. The results suggest that every aspect from successful identification of the genome component of the virus, timely detection of outbreak, monitoring, containment, risk assessment, prediction, development of vaccines to dissemination of knowledge and information related to COVID-19 is targeted through use of digital tools. Emerging technologies like Artificial Intelligence, Robotics, Artificial Intelligence of Things (AIoT), Blockchain are the focus of innovations. Telemedicine is being used increasingly for consulting. Migrant population, lack

of bandwidth, infrastructure, lack of scalability and sustainability of the digital solutions, unavailability of timely information are some of the challenges to adoption. Despite the barriers and challenges, there is still a vast opportunity for the leveraging technology to combat COVID-19 spread in the country with National Digital Health Mission (NDHM), Atma Nirbhar Bharat Scheme and booming entrepreneurship models. Moreover the learnings from COVID should be utilized for improving health outcomes in the post-pandemic era.

Date 17-19 March 2021

Extra Journal Abbreviation: 2021 8th International Conference on Computing for Sustainable Global Development (INDIACom)

Pages 244-249

Proceedings Title 2021 8th International Conference on Computing for Sustainable Global Development (INDIACom)

Conference Name 2021 8th International Conference on Computing for Sustainable Global Development (INDIACom)

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Tags:

Blockchain, Technological innovation, Robotics, Telemedicine, Government, Artificial Intelligence of Things, COVID-19, Digital Health Technologies, Electronic healthcare, Emerging Technologies in Healthcare, NDHM, Pandemics, Post Pandemic Digital Health, Tools

Dissociation Between Users' Explicit and Implicit Attitudes Toward Artificial Intelligence: An Experimental Study

Item Type Journal Article

Author V. Fietta

Author F. Zecchinato

Author B. D. Stasi

Author M. Polato

Author M. Monaro

Abstract The latest developments in the field of artificial intelligence (AI) have given rise to many ethical and socio-economic concerns. Nonetheless, the impact of AI technologies is evident and tangible in our everyday life. This dichotomy leads to mixed feelings toward AI: people recognize the positive impact of AI, but they also show concerns, especially about their privacy and security. In this article, we try to understand whether the implicit and explicit attitudes toward AI are coherent. We investigated explicit and implicit attitudes toward AI by combining a self-report measure and an implicit measure, i.e., the implicit association test. We analyzed the explicit and implicit responses of 829 participants. Results revealed that while most of the participants explicitly express a positive attitude toward AI, their implicit responses seem to point in the opposite direction. Results also show that, in both the explicit and implicit measures, females show a more negative attitude than males, and people who work in the field of AI are inclined to be positive toward AI.

Date June 2022

DOI 10.1109/THMS.2021.3125280**Volume** 52**Pages** 481-489**Publication** IEEE Transactions on Human-Machine Systems**ISSN** 2168-2305**Issue** 3**Journal Abbr** IEEE Transactions on Human-Machine Systems**Date Added** 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

Artificial intelligence, Task analysis, Robots, Production, Cognition, Atmospheric measurements, Particle measurements, Attitudes toward artificial intelligence (AI), explicit–implicit cognition, implicit association test (IAT), implicit attitudes

DRIVE: Dependable Robust Interpretable Visionary Ensemble Framework in Autonomous Driving

Item Type Conference Paper**Author** S. Lai**Author** T. Xue**Author** H. Xiao**Author** L. Hu**Author** J. Wu**Author** N. Feng**Author** R. Guan**Author** H. Liao**Author** Z. Li**Author** Y. Yue

Abstract Recent advancements in autonomous driving have seen a paradigm shift towards end-to-end learning paradigms, which map sensory inputs directly to driving actions, thereby enhancing the robustness and adaptability of autonomous vehicles. However, these models often sacrifice interpretability, posing significant challenges to trust, safety, and regulatory compliance. To address these issues, we introduce DRIVE – Dependable Robust Interpretable Visionary Ensemble Framework in Autonomous Driving, a comprehensive framework designed to improve the dependability and stability of explanations in end-to-end unsupervised autonomous driving models. Our work specifically targets the inherent instability problems observed in the Driving through the Concept Gridlock (DCG) model, which undermine the trustworthiness of its explanations and decisionmaking processes. We define four key attributes of DRIVE: consistent interpretability, stable interpretability, consistent output, and stable output. These attributes collectively ensure that explanations remain reliable and robust across different scenarios and perturbations. Through extensive empirical evaluations, we demonstrate the effectiveness of our framework in enhancing the stability and dependability of explanations, thereby addressing the limitations of

current models. Our contributions include an in-depth analysis of the dependability issues within the DCG model, a rigorous definition of DRIVE with its fundamental properties, a framework to implement DRIVE, and novel metrics for evaluating the dependability of concept-based explainable autonomous driving models. These advancements lay the groundwork for the development of more reliable and trusted autonomous driving systems, paving the way for their broader acceptance and deployment in real-world applications. “We can only see a short distance ahead, but we can see plenty there that needs to be done.” – Alan Turing

Date 19-23 May 2025

DOI 10.1109/ICRA55743.2025.11127884

Extra Journal Abbreviation: 2025 IEEE International Conference on Robotics and Automation (ICRA)

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Tags:

Autonomous vehicles, Safety, Robustness, Standards, Measurement, Analytical models, Perturbation methods, Power system stability, Robotics and automation, Stability criteria

Enhanced Machine Learning Algorithms for Validating the Biometrics in VANET

Item Type Conference Paper

Author P. Chithaluru

Author L. Jena

Author B. Patra

Author N. Panda

Abstract In this rapidly sprouting technological era, trans-portion is also becoming intelligent and driverless. Ad hoc networking plays an essential role in making the vehicle to vehicle and vehicle to infrastructure communication environment compatible with all types of users. As technology grows, the risk of security and privacy also grows side by side. However, the intruders can easily find out holes in the security network and gather information from them. The scope of the paper is to propose a methodology that inflates the security of stored data in the hardware of vehicular networks and identity verification of vehicles in communication easier. The importance of the present research is that these proposed simulation techniques help in maintaining the security of temper proof devices used in VANETs for the storage of private data. Authors communicate all sensitive data like bank details, vehicle details and personal details over wireless networks during communication. Biometric data and many other cryptographic techniques are used to secure the communication. Proposed methodologies are very efficient, less time consuming and less cost effective too which can be used in any type of intelligent vehicle with no extra hardware requirement.

Date 5-6 Aug. 2022
DOI 10.1109/MLCSS57186.2022.00011
Extra Journal Abbreviation: 2022 International Conference on Machine Learning, Computer Systems and Security (MLCSS)
Pages 11-16
Proceedings Title 2022 International Conference on Machine Learning, Computer Systems and Security (MLCSS)
Conference Name 2022 International Conference on Machine Learning, Computer Systems and Security (MLCSS)
Date Added 20/09/2026, 7:46:49 pm
Modified 20/09/2026, 7:46:49 pm

Tags:

Security, Vehicular ad hoc networks, Privacy, Machine learning, Machine learning algorithms, Wireless networks, Biometrics, Intelligent vehicles, Biometrics (access control), Cryptographic technique, Vehicular network

Enhancing Privacy in 6G Networks with Federated Learning for Data Analytics

Item Type Conference Paper
Author H. Gao
Author Y. Yu
Author Z. Pang
Abstract This paper proposes a federated learning-based framework specifically designed for privacy-preserving data analytics in 6G networks. Federated learning enables data to remain local on edge devices while only transmitting model updates, ensuring sensitive user data is not exposed. Our experimental results demonstrate that this framework not only maintains high privacy standards but also improves the speed and accuracy of data analytics when compared to traditional centralized methods. The proposed framework addresses the privacy challenges of handling vast, heterogeneous data in 6G environments. By enhancing privacy preservation and processing efficiency, federated learning is positioned as a critical enabler for next-generation networks.
Date 27-30 Oct. 2025
DOI 10.1109/WF-IoT64238.2025.11270609
Extra Journal Abbreviation: 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)
ISBN 2768-1734
Pages 1-6
Proceedings Title 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)
Conference Name 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)
Date Added 20/09/2026, 7:46:49 pm
Modified 20/09/2026, 7:46:49 pm

Tags:

Data privacy, Privacy, Federated learning, Heuristic algorithms, privacy preservation, Real-time systems, Standards, Accuracy, 6G mobile communication, 6G networks, Data analysis, data analytics, decentralized computing, federated learning, Next generation networking

Enhancing Real-time Facial Sentiment Recognition Systems through Deep Learning Architectures

Item Type Conference Paper

Author K. Geetha Rani

Author R. Selvaraj

Author A. Swaminathan

Author R. Surendran

Abstract Facial Emotion Recognition (FER) has emerged as a pivotal domain within computer vision, with applications spanning diverse fields such as human-computer interaction, mental health monitoring, and security systems. While significant strides have been made in recent years, accurate recognition of nuanced emotions, particularly fear, remains a challenging task. This research contributes to the advancement of FER technology, paving the way for more sophisticated and empathetic human-computer interactions. By discussing about the reliable and efficient methods for real-time fear detection, this study enables the potential to revolutionize various applications, including mental health monitoring, security surveillance, and autonomous vehicles.

Date 23-25 Oct. 2024

DOI 10.1109/ICSSAS64001.2024.10760738

Extra Journal Abbreviation: 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)

Pages 519-524

Proceedings Title 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)

Conference Name 2024 2nd International Conference on Self Sustainable Artificial Intelligence Systems (ICSSAS)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Security, Mental health, Human computer interaction, Transfer learning, data preprocessing, effectiveness, EMOTIC dataset, Emotion recognition, Face recognition, facial emotion recognition, Facial sentiment recognition, machine learning baselines, Surveillance, Real-time systems, Accuracy, Videos

Enhancing Road Safety: A Comparative Study Between UAV-Assisted and Autonomous Vehicles

Item Type Conference Paper**Author** S. Bouassida**Author** L. Nouveliere**Author** N. Neji**Author** J. Neji

Abstract The integration of connected autonomous vehicles (CAV) on open roads has gained significant progress in addressing road safety. These vehicles use advanced sensor technology to perceive and react to the road environment, reducing accident risks. Despite these advancements, limitations persist in their perception capabilities. To overcome these limitations, interest is growing in using Unmanned Aerial Vehicles (UAVs) for traffic surveillance, offering extensive coverage and enhanced responsiveness over fixed sensors. In this article, by tackling an optimization problem in road safety using Particle Swarm Optimization (PSO), we particularly focus on a situation where a random flow of vehicles aims to navigate an intersection safely. We compare two scenarios with and without the assistance of an UAV: one where vehicles autonomously manage their speed, and another one where an UAV improve traffic management. Simulation results underscore the pivotal role of drone-assisted vehicles in enhancing road safety, compared to sensors embedded within CAVs. Towards the end of the article, we explore the efficiency of the drone strategy by addressing the issue of delay in the acceptance and implementation of optimal speed instructions, comparing scenarios with and without this delay.

Date 12-15 Dec. 2024**DOI** 10.1109/ICARCV63323.2024.10821584**Extra** Journal Abbreviation: 2024 18th International Conference on Control, Automation, Robotics and Vision (ICARCV)**ISBN** 2474-963X**Pages** 1118-1123**Proceedings Title** 2024 18th International Conference on Control, Automation, Robotics and Vision (ICARCV)**Conference Name** 2024 18th International Conference on Control, Automation, Robotics and Vision (ICARCV)**Date Added** 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

UAV, Safety, Acceptability, CAV, drone to vehicle communication (U2V), Drones, Multi-Agent system, optimization, road safety, Road safety, Simulation, Traffic control, Surveillance, Delays, Autonomous aerial vehicles, Robot sensing systems, Stability analysis

Ethical Concerns With Regards to Artificial Intelligence: A National Public Poll in Taiwan

Item Type Journal Article**Author** W. Li-Yun Chang**Author** S. -Y. Liu

Author Y. -K. Liao

Author T. S. -H. Lee

Abstract Ethical concerns about how artificial intelligence (AI) impacts individuals and society are increasing rapidly, but few studies have systematically investigated the public awareness of AI ethics. This research collected and analyzed data from a public poll in Taiwan, an Asian region with a developed economy and specific social conditions, to identify societal views on AI ethics. The analysis of 84 AI ethics guidelines worldwide provided the survey framework covering five ethical principles: transparency, fairness, privacy, nonmaleficence, and accountability. The overarching goal was to determine the commonalities and differences in the ethical concerns of Taiwanese laypersons toward AI. Participants aged from 20 to 70 ($N=1,200$) completed a computer-assisted random-digit-dial telephone survey, which utilized ethical scenarios to capture social views, and item validity was confirmed using focus-group interviews. Results found that respondents concerned about nonmaleficence the most, emphasizing that AI applications should not harm humans. Taiwanese people therefore tended to support strict AI technology regulation. It was particularly interesting that different patterns of public concern emerged about accountability, with the opinions on attributing responsibility to stakeholders varying with scenarios and the public's backgrounds. Those with higher education levels tended to attribute more responsibility to the industry, whereas those who had only received elementary-school education attributed accountability to AI developers. For self-driving cars, accountability was attributed to AI developers, whereas for medical decision-making, the accountability was attributed to the hospitals. These findings may help to elucidate the associations between societal views and the ethical principles of AI worldwide.

Date 2024

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Pages 133595-133605

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Journal Abbr IEEE Access

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Tags:

Artificial intelligence, Privacy, public perception, Regulation, Surveys, ethical concerns, ethical principles, ethics, society, Stakeholders, Ethics, Guidelines

Event-Triggered Federated Reinforcement Learning for Resource Allocation in NR-V2X

Item Type Conference Paper

Author C. Jin

Author B. -J. Hu

Abstract Resource allocation has a significant impact on the efficiency of vehicle-to-everything (V2X) networks. In order to achieve efficient resource allocation, this paper presents a novel Event-Triggered Federated Reinforcement Learning (ETFRL) scheme. This scheme modifies the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) scheme to enable agents to be aware of global interference, thereby optimizing overall network performance. Furthermore, the model integrates Federated Learning (FL) to protect user data privacy and accelerate training speed. It also incorporates an Event-Triggered mechanism to selectively initiate the Deep Reinforcement Learning (DRL) process, reducing the training burden. Simulation results demonstrate that the ETFRL scheme achieves near-equivalent performance to the time-triggered algorithm while significantly reducing execution times. This indicates that the ETFRL scheme is a promising approach for enhancing spectrum efficiency in V2X resource allocation problem.

Date 27-30 Oct. 2025

DOI 10.1109/WF-IoT64238.2025.11270684

Extra Journal Abbreviation: 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

ISBN 2768-1734

Pages 1-6

Proceedings Title 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

Conference Name 2025 IEEE 11th World Forum on Internet of Things (WF-IoT)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Data privacy, V2X, Simulation, Deep reinforcement learning, Federated learning, Internet of Things, Vehicle-to-everything, Resource management, Training, Deep Reinforcement Learning, Event detection, Federal Learning, Interference, Platoon cooperation, Resource Allocation

Explainable Edge Computing in a Distributed AI - Powered Autonomous Vehicular Networks

Item Type Conference Paper

Author P. Mahajan

Author G. S. Aujla

Author C. R. Krishna

Abstract In the ever-dynamic landscape of autonomy and un-paralleled intelligence, the vehicles moving across any geospatial terrain create an Artificial Intelligence (AI)-powered vehicular network that can interact quickly to make autonomous decisions. However, the efficient working of autonomous vehicles (AV s) relies on seamless data integration from various resources, including Vehicle-to-vehicle (V2V), Vehicle-to-Infrastructure (V2I), and other communications in the AI-powered networks. In such scenarios, the velocity and pattern of vehicle mobility and motion require high effectiveness and efficiency in AI-enabled decision-making. Additionally, understanding the critical basis or feature data that triggers a particular decision from an AI model is also essential to enhance user acceptance and trust. This paper introduces an explainable edge computing-based

solution to enhance the performance of AVs in an AI-powered vehicular network. The solution focuses on harnessing the power of local edge through Explainable Artificial Intelligence (XAI). This data is sourced from local edge computing nodes and is efficiently disseminated to the global edge. A comprehensive approach is designed for identifying and amalgamating important feature data from the local edge to the global edge. Through this cooperative approach, autonomous vehicular networks attain elevated efficiency, accuracy, and adaptability, making the driving experience of an AV more reliable and secure.

Date 9-13 June 2024

DOI 10.1109/ICCWorkshops59551.2024.10615824

Extra Journal Abbreviation: 2024 IEEE International Conference on Communications Workshops (ICC Workshops)

ISBN 2694-2941

Pages 1195-1200

Proceedings Title 2024 IEEE International Conference on Communications Workshops (ICC Workshops)

Conference Name 2024 IEEE International Conference on Communications Workshops (ICC Workshops)

Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

Autonomous Vehicles, Vehicular ad hoc networks, Decision making, Vehicular Networks, Edge Computing, Data integration, Explainable AI, Conferences, Data models, Vehicle-to-infrastructure, Explainable Artificial Intelligence

Exploring Artificial Intelligence (AI) Impact on Businesses: Perspectives from Big Data and Security

Item Type Conference Paper

Author S. Gopalkrishnan

Author A. Reddipogu

Abstract The concept of Artificial Intelligence (AI) has changed with time, and with the advent of Big Data, Machine Learning, Deep Learning, and Artificial Neural Networks, the fundamentals of AI and the way it is perceived in today's life have undergone a sea-change. However, along with this, the perception of Data and Security has also come a long way. Security earlier meant being in a secure, safe, risk-free zone; it has now metamorphosed itself into a concept of perimeter-driven layers of security – which changes from person to person. Against this backdrop, some of the pertinent questions that face us today are: Does AI aim to replace Human Intelligence? Would AI overtake Human Intelligence? Can AI 'learn' on its own – thereby teaching itself new concepts and mechanisms – while on the fly? This article takes us through the various concepts of AI, its types and modes of growth and expected future development, leading to its impact on society, businesses, culture and people. Further, this study leads to impact points of AI development and progress on points like data and security and possible costs of security and whether they would be restricted to financial costs alone.

Towards the end, the authors put forth thoughts to ponder on whether Big Data and AI are the answers to all questions that lie ahead; and how AI can play a role in the betterment of Data and Security available to global citizens of future societies; with a responsible impact on businesses, economies, people and culture.

Date 26-27 Jan. 2023

DOI 10.1109/CyMaEn57228.2023.10051065

Extra Journal Abbreviation: 2023 International Conference On Cyber Management And Engineering (CyMaEn)

Pages 12-17

Proceedings Title 2023 International Conference On Cyber Management And Engineering (CyMaEn)

Conference Name 2023 International Conference On Cyber Management And Engineering (CyMaEn)

Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

Security, Deep learning, Education, Costs, Big Data, Human intelligence, Artificial Intelligence Experiences, Artificial neural networks, Big Data vs Small Data, Business Impact of AI, Data and Security

Extraction and Analysis of Highway On-Ramp Merging Scenarios from Naturalistic Trajectory Data

Item Type Conference Paper

Author L. Klitzke

Author K. Gimm

Author C. Koch

Author F. Köster

Abstract Automated Vehicles are envisioned to transform the future industrial and private transportation sectors. However, due to the system's enormous complexity, functional verification and validation of safety aspects are essential before the technology merges into the public domain. In recent years, a scenario-driven approach has gained acceptance, emphasizing the requirement of a solid data basis of scenarios. The large-scale research facility Test Bed Lower Saxony (TFNDS) of the German Aerospace Center (DLR) enables the provision of ample information for a database of scenarios on highways. For that purpose, however, the scenarios of interest must be identified and extracted from the collected Naturalistic Trajectory Data (NTD). This work addresses this problem and proposes a methodology for on-ramp scenario extraction, enabling scenario categorization and assessment. A Hidden Markov Model and Dynamic Time Warping is utilized for extraction and a decision tree with the Surrogate Measure of Safety Post Encroachment Time for categorization and assessment. The efficacy of the approach is shown with a dataset of NTD collected on the TFNDS.

Date 8-12 Oct. 2022

DOI 10.1109/ITSC55140.2022.9922191

Extra Journal Abbreviation: 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)
Pages 654-660
Proceedings Title 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)
Conference Name 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)
Date Added 20/09/2026, 7:46:49 pm
Modified 20/09/2026, 7:46:49 pm

Tags:

Safety, Transportation, Connected and Automated Vehicles, Time measurement, Road transportation, Transforms, Highway On-Ramp Merging, Merging, Naturalistic Trajectory Data, Scenario Extraction, Trajectory

Facilitating Research for IoT-Based Vehicle-to-Everything (V2X) Survey Web Application Design

Item Type Conference Paper
Author Z. Wang
Author B. Hu
Author Y. Gao
Author S. Huang
Author Y. Dong
Abstract With the development of the Internet of Things (IoT) related applications, multiple requirements and massive raw data will be generated accordingly. How to address the requirements for versatile survey creation and participation across diverse linguistic environments still is a challenge. This paper aims to propose a potential analysis method to distinguish multiple types of news on social media while facilitating user engagement and relevant data collection by a cloud-based Fakenewz web application. Based on the integration of various question types, multilingual support, and graphical data representation, Fakenewz caters to the evolving demands of survey production and feedback gathering, which enhances user experience, and the effectiveness of multilingual support in survey deployment. The results emphasise a strengthened foundation, improved user interaction, and innovation within the online survey landscape.
Date 29-31 Dec. 2023
DOI 10.1109/SMC-IoT62253.2023.00023
Extra Journal Abbreviation: 2023 2nd International Conference on Sensing, Measurement, Communication and Internet of Things Technologies (SMC-IoT)
Pages 85-90
Proceedings Title 2023 2nd International Conference on Sensing, Measurement, Communication and Internet of Things Technologies (SMC-IoT)
Conference Name 2023 2nd International Conference on Sensing, Measurement, Communication and Internet of Things Technologies (SMC-IoT)
Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

User experience, Technological innovation, Agile methodology, Fakenewz, IoT, Online survey platform, Rank Order, Slider, Social networking (online), Data collection, Surveys, Production, Linguistics

Federated Learning for Image Captioning: A Comprehensive Review of Privacy-Preserving Collaborative Model Training in Distributed Environments

Item Type	Conference Paper
Author	R. Padate
Author	M. Kalla
Author	A. Gupta
Author	A. Sharma
Abstract	This study presents a comprehensive review of the use of federated learning in the context of image captioning in distributed environments. It focuses on key aspects such as privacy preservation, data locality, and collaborative model training. The evolution of federated learning and its unique characteristics are explored, along with an examination of available open-source frameworks specific to image captioning. The study categorizes different approaches to federated learning for image captioning and showcases recent applications in diverse domains, including medical imaging, edge computing, autonomous vehicles, social media, and cross-domain image analysis. Additionally, optimization techniques, security analysis, and research challenges are discussed, encompassing data heterogeneity, privacy preservation, communication efficiency, limited labeling, scalability, and robustness against adversarial attacks. This comprehensive review contributes to a deeper understanding of federated learning for image captioning and highlights areas for further research and advancement in the field.
Date	19-21 July 2023
DOI	10.1109/ICECAA58104.2023.10212161
Extra	Journal Abbreviation: 2023 2nd International Conference on Edge Computing and Applications (ICECAA)
Pages	857-865
Proceedings Title	2023 2nd International Conference on Edge Computing and Applications (ICECAA)
Conference Name	2023 2nd International Conference on Edge Computing and Applications (ICECAA)
Date Added	20/09/2026, 7:46:49 pm
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Tags:

Data privacy, Social networking (online), Federated learning, Privacy preservation, Data Privacy, Data Security, Federated Learning, Image Captioning, Collaboration, Scalability, Training, Analytical models

First Responder Communication Systems: A Technical Analysis of Interoperability and Situational Awareness Optimization

Item Type Conference Paper

Author S. Majumdar

Author A. Awasthi

Author S. E. Kirkley

Abstract This research examines the critical challenges and technical requirements of communication systems for first responders, including law enforcement, fire services, and emergency medical services (EMS). It identifies limitations in current systems, such as interoperability barriers, outdated radio infrastructure, and inadequate real-time data integration. Emerging technologies, including IoT-enabled autonomous vehicles (AVs), 5G communication networks, and Vehicle-to-Everything (V2X) protocols, are analyzed for their potential to address these issues. These advancements offer enhanced real-time geolocation accuracy, improved situational awareness, and seamless multi-agency information exchange. The findings highlight key technical obstacles and provide insights into how innovative solutions can enhance the operational efficiency, reliability, and safety of first responder communication systems.

Date 6-8 Jan. 2025

DOI 10.1109/CCWC62904.2025.10903555

Extra Journal Abbreviation: 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)

Pages 00331-00339

Proceedings Title 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)

Conference Name 2025 IEEE 15th Annual Computing and Communication Workshop and Conference (CCWC)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Law enforcement, Reliability, Internet of Things, Public security, Real-time systems, Protocols, Testing, 5G mobile communication, Medical services, 5G Networks, Advanced Communication Systems, Emergency Response, First Responder Communication, Geolocation Tracking, Interoperability, IoT Devices, Public Safety Communications, Public Safety Operations, Situational Awareness

Foundation Models: From Current Developments, Challenges, and Risks to Future Opportunities

Item Type Conference Paper

Author A. Hussain

Author S. Ali

Author U. E. Farwa

Author M. A. I. Mozumder

Author H. -C. Kim

Abstract Foundation models are a revolutionary technology in the field of artificial intelligence that could usher in an era of transformation in every field with its vision and language understating capabilities while maintaining remarkable performance. Here, we extensively review state-of-the-art foundation models in various fields such as LLMs, GPT, BERT, CLIP, etc. We also demonstrate some famous areas where foundation models are gaining much popularity, such as general foundation models, foundation models in the medical domain, education, law and finance, mathematics, autonomous driving, etc. These foundation models are trained on large-scale data by leveraging the capabilities of self-supervised learning approaches that ensure outperforming accuracy for relevant downstream tasks. They are successful due to pioneering architectural innovations, especially synergistically interleaving transformers and convolutional neural networks. These models have exhibited adaptability and resilience in various data patterns and conditions through sophisticated training paradigms such as self-supervised and supervised learning methods. While they hold transformative potential, they also have many challenges. The scale and quality of training data are still important predictors of model performance, and the need for interpretable and explainable AI systems becomes critically important. We highlight important research opportunities, high-performing computationally efficient and scalable architectures, and approaches to enable multimodal learning capacities. This review sheds light on the current state of foundation models and proposes a roadmap for their successful transitions.

Date 16-19 Feb. 2025

DOI 10.23919/ICACT63878.2025.10936649

Extra Journal Abbreviation: 2025 27th International Conference on Advanced Communications Technology (ICACT)

ISBN 1738-9445

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Proceedings Title 2025 27th International Conference on Advanced Communications Technology (ICACT)

Conference Name 2025 27th International Conference on Advanced Communications Technology (ICACT)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Technological innovation, Computer architecture, Computational efficiency, Reviews, Foundation models, Computational modeling, Data models, Training, Training data, digital pathology foundation models, LLMs, self-supervised learning, Self-supervised learning, vision-language models

Framework for planning the mobility service based on autonomous vehicles

Item Type	Conference Paper
Author	D. Földes
Author	C. Csiszár
Abstract	Autonomous vehicles (AVs) facilitate alteration in the passenger transportation. Shared, demand-driven mobility services based on small-capacity AVs emerge in urban areas. In order to provide highly personalized services and to enhance the acceptance of AVs, the user expectations are to be revealed. The research question is what kind of information is needed to plan various service types with the consideration of user expectations and operational constraints. Accordingly, the planning functions, as well as their input and output data have been identified. Then, the connections between the functions have been explored. In order to provide input data for planning functions, we have elaborated a data collection method applying a questionnaire survey with stated preferences. Based on the survey data correlations could be revealed between socio-demographic characteristics or mobility habits and expected mobility service attributes. The results can be used as a framework for planning mobility services based on AVs.
Date	24-25 May 2018
DOI	10.1109/SCSP.2018.8402651
Extra	Journal Abbreviation: 2018 Smart City Symposium Prague (SCSP)
Pages	1-6
Proceedings Title	2018 Smart City Symposium Prague (SCSP)
Conference Name	2018 Smart City Symposium Prague (SCSP)
Date Added	20/09/2026, 7:46:50 pm
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Tags:

autonomous vehicles, questionnaire survey, Automobiles, Information systems, Personnel, Public transportation, Legged locomotion, Planning, service planning, stated preference, user expectation

From 5G to 6G: The Role of AI, Machine Learning, and Deep Learning in Wireless Systems

Item Type	Conference Paper
Author	P. M. Gote
Author	P. Kumar
Author	P. Verma
Author	P. Yesankar
Author	A. Pawar
Author	S. Saratkar
Abstract	The swift progression of wireless communication technologies-specifically from 5G to 6G is an approach that could be the most significant revolutionary leap towards changing connectivity and data transmission forever. Additionally, algorithms of machine learning (ML) and deep learning (DL) in 5G and 6G can

offer promising advancements toward predictive maintenance, dynamic spectrum management, and enhanced security. The review of the potentially transformative nature of these technologies underlines the need for future directions of research efforts called for in circumventing challenges towards making full capability exploitation of 6G possible.

Date	18-20 Feb. 2025
DOI	10.1109/ICSADL65848.2025.10933016
Extra	Journal Abbreviation: 2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)
Pages	447-452
Proceedings Title	2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)
Conference Name	2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)
Date Added	20/09/2026, 7:46:50 pm
Modified	20/09/2026, 7:46:50 pm

Tags:

Security, Deep learning, Sentiment analysis, Machine learning, Technological innovation, Reviews, Heuristic algorithms, Wireless communication, 5G mobile communication, 6G mobile communication, 5G technology, 6G technology, AI integration, AI RAN alliance (NVIDIA), Cellular communication, Tactile Internet

From Principles to Practice: Responsible Implementation of Emerging Technologies Through Innovation Dilemmas

Item Type	Journal Article
Author	C. Carbajal-Pina
Author	N. Acur
Abstract	This article explores the imperative role of responsible innovation (RI) in guiding the development and integration of emerging technologies within society. With technological progress rapidly advancing, it becomes ever more imperative to confront the complex ethical, social, economic, and environmental challenges that accompany these innovations. This research highlights the significance of RI practices, advocating for a principle-based approach to technology development. It asserts that such an approach, which is rooted in societal values and ethical standards, is crucial for nurturing long-term societal well-being and fostering trust in new technologies, as opposed to relying solely on rule-based technology development governed by regulations. Our analysis begins with the definition of each RI principle, followed by an exploration of its relevance in addressing innovation dilemmas. Our case studies involve various sectors, including artificial intelligence, connected autonomous vehicles, and quantum computing, to showcase the practical application of these principles in real-world contexts. Through these instances, we highlight the efficacy of RI in both mitigating risks and maximizing the benefits of new technologies. In addition, the article investigates the dynamics of implementing RI within organizational frameworks and innovation processes. It provides strategic recommendations for technology managers and policymakers on how to

operationalize RI principles effectively, emphasizing the need for an adaptive approach that is responsive to the rapid changes in the technology landscape. The study contributes to the academic discourse by highlighting that proactive engagement with RI can lead to more sustainable and socially acceptable technology outcomes. Ultimately, this research underscores the importance of responsible practices in innovation, proposing that a principled approach to technology development is essential for fostering long-term societal well-being and trust in new technologies.

Date 2025

DOI 10.1109/TEM.2024.3479413

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Journal Abbr IEEE Transactions on Engineering Management

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Tags:

Regulation, Technological innovation, Uncertainty, Navigation, emerging technologies, Quantum computing, Socioeconomics, Faces, Ethics, Standards, Business, Connected autonomous vehicles (CAV), innovation dilemmas, principles-based innovation, quantum computing, responsible innovation (RI)

Get in Line: Ongoing Co-presence Verification of a Vehicle Formation Based on Driving Trajectories

Item Type Conference Paper

Author C. Vaas

Author M. Juuti

Author N. Asokan

Author I. Martinovic

Abstract Intelligent transportation systems and the advent of smart cities have created a renewed research interest in vehicular networks (VANET). These ad-hoc networks are the key technology for new collaborative approaches to increase the efficiency and safety of our roads. In effect, city-scale field trials are being conducted by major high-tech companies to explore the capabilities and limitations of vehicle-to-infrastructure and vehicle-to-vehicle communication. Initial advances have led to safety enhancing applications like the electronic emergency brake light, cooperative collision avoidance and cooperative adaptive cruise control. In IEEE standard 1609.2, security measures to guarantee the integrity and authenticity of VANET messages are specified. However, physical properties like spatial proximity and driving direction are not considered. These become notably important when vehicles make decisions that concern the safety of users for example to avoid a collision. We propose a novel approach to verify the ongoing copresence of two vehicles. Our method is based on the observation

that the trajectory through a road network can be used to uniquely define a vehicle's location as well as its driving direction. Our system provides a protocol to authenticate VANET messages for a group of vehicles driving in succession and to de-authenticate vehicles that have left the formation. To demonstrate the feasibility of trajectories as proof for co-presence, we implemented a smartphone application and conducted driving experiments under real-world conditions. We analyze the road network of several major cities from different continents to show the generalizability of our approach. Additionally, we systematically evaluate the security properties of our system by performing city-scale simulations under realistic conditions.

Date 24-26 April 2018

DOI 10.1109/EuroSP.2018.00022

Extra Journal Abbreviation: 2018 IEEE European Symposium on Security and Privacy (EuroS&P)

Pages 199-213

Proceedings Title 2018 IEEE European Symposium on Security and Privacy (EuroS&P)

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Modified 20/09/2026, 7:46:50 pm

Tags:

Safety, Security, Vehicular ad hoc networks, Privacy, VANET, Roads, Protocols, Trajectory, co-presence verification, platooning, road train

GoComfort: Comfortable Navigation for Autonomous Vehicles Leveraging High-Precision Road Damage Crowdsensing

Item Type Journal Article

Author L. Chen

Author X. He

Author X. Zhao

Author H. Li

Author Y. Huang

Author B. Zhou

Author W. Chen

Author Y. Li

Author C. Wen

Author C. Wang

Abstract Recent years have witnessed rapid advances in autonomous driving technologies. Autonomous vehicles are more likely to be accepted if they drive comfortably to avoid potholes, bumps, and other road damage conditions, especially when there are elderly and disabled passengers on board. Traditionally, sensing road damage conditions is either labor-intensive by field investigation and reporting, or inaccurate by surveillance cameras and driving recorders due to limited perspective. In this paper, we propose GoComfort, a crowdsensing-based framework to provide low-cost and fine-grained comfortable navigation for autonomous vehicles with road damage

identification leveraging high-precision road sensing data. First, we propose to exploit city-wide autonomous vehicle fleets as crowdsensing participants, and employ an edge-cloud-hybrid computing paradigm to efficiently collect high-precision road damage-related data, including 3D LiDAR point clouds and street view images. Second, we design an accurate road damage identification model fusing spatial structures of point clouds and texture features of street view images, and use an active learning-based method to address the sparse labels issue. Finally, we devise two comfortable navigation scenarios, i.e., fine-grained road damage avoidance and coarse-grained city-wide navigation, and propose a hierarchical road damage assessment diagram for comfortable route planning. Experiments using real-world road sensing data in Xiamen, China show that our approach identifies road damage conditions with an accuracy of 87.5%, and achieves a user acceptance rate of 93.8% in riding comfort evaluation, outperforming the state-of-the-art baselines.

Date 1 Nov. 2023

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Tags:

Autonomous vehicles, Urban areas, Comfortable route planning, Crowdsensing, mobile crowdsensing, Navigation, Point cloud compression, road damage identification, urban computing, Sensors, Roads

How Do Starship Robots Affect Everyday Campus Life? An Exploratory Posting Board Analysis and Interview-Based Study

Item Type Conference Paper

Author A. Schneider

Author A. Robinson

Author C. Grimm

Author N. T. Fitter

Abstract The rapid emergence of food delivery robots in public spaces has raised important questions regarding public perceptions and policy creation. One method for addressing these questions is examining the relationship between delivery robots and the communities they already serve. We assessed our university community's experiences with, and perceptions of, the Starship Technologies robots (robots that currently operate on campus) using two efforts: analysis of online posting board content related to the robots and interviews of campus community members about the robots. Perspectives captured in the online post analysis tended to be negative, while views tended to be positive in the interview results. At the same time, both results showed differing opinions and complexity; one tension that emerged in both efforts, for example, is the potential of the robots to

both benefit and impede disability communities on campus. Further, there were fundamental misunderstandings about what data the robots can and do record. This research can help to inform roboticists and policymakers whose work relates to autonomous robots in public spaces.

Date 26-30 Aug. 2024

DOI 10.1109/RO-MAN60168.2024.10731287

Extra Journal Abbreviation: 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)

ISBN 1944-9437

Pages 528-534

Proceedings Title 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)

Conference Name 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN)

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Tags:

Complexity theory, Robots, Interviews, Autonomous robots

“How technical do you get? I’m an English teacher”: Teaching and Learning Cybersecurity and AI Ethics in High School

Item Type Conference Paper

Author Z. Kilhoffer

Author Z. Zhou

Author F. Wang

Author F. Tamton

Author Y. Huang

Author P. Kim

Author T. Yeh

Author Y. Wang

Abstract Today’s cybersecurity and AI technologies are often fraught with ethical challenges. One promising direction is to teach cybersecurity and AI ethics to today’s youth. However, we know little about how these subjects are taught before college. Drawing from interviews of US high school teachers (n=16) and students (n=11), we find that cybersecurity and AI ethics are often taught in non-technical classes such as social studies and language arts. We also identify relevant topics, of which epistemic norms, privacy, and digital citizenship appeared most often. While teachers leverage traditional and novel teaching strategies including discussions (treating current events as case studies), gamified activities, and content creation, many challenges remain. For example, teachers hesitate to discuss current events out of concern for appearing partisan and angering parents; cyber hygiene instruction appears very ineffective at educating youth and promoting safer online behavior; and generational differences make it difficult for teachers to connect with students. Based on the study results, we offer practical suggestions for educators, school administrators, and cybersecurity

practitioners to improve youth education on cybersecurity and AI ethics.

Date	21-25 May 2023
DOI	10.1109/SP46215.2023.10179333
Extra	Journal Abbreviation: 2023 IEEE Symposium on Security and Privacy (SP)
ISBN	2375-1207
Pages	2032-2032
Proceedings Title	2023 IEEE Symposium on Security and Privacy (SP)
Conference Name	2023 IEEE Symposium on Security and Privacy (SP)
Date Added	20/09/2026, 7:46:50 pm
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Tags:

AI ethics, Artificial intelligence, cybersecurity, Privacy, Education, Computer crime, education, Behavioral sciences, Ethics, Art

How Trustworthy are Over-The-Air (OTA) Updates for Autonomous Vehicles (AV) to Ensure Public Safety?: A Threat Model-based Security Analysis

Item Type	Conference Paper
Author	N. M. Istiak Chowdhury
Author	R. Hasan
Abstract	The rise of autonomous vehicles (AV) has transformed our transportation sectors in recent years. The evolution of technology has made AVs more popular among users. The Over-The-Air (OTA) updates for AVs are widely adopted by the AV industry. These updates are used to deliver improved functionality and security patches to ensure public safety. However, this system is susceptible to security threats since the wireless communication link between the car, the manufacturer, and the cloud-based web server is inherently insecure. While there are studies that explored multiple cybersecurity frameworks and tools for OTA updates, many challenges remain unaddressed. To build a robust trustworthy system, we need to perform a thorough security analysis by creating a threat model. In this study, we use the CIAA and STRIDE threat modeling process to analyze the OTA updates security. Our analysis demonstrates numerous dangers and vulnerabilities. To increase the security of OTA updates, we also suggest several corrective mitigation steps.
Date	14-15 May 2024
DOI	10.1109/WFPST58552.2024.00025
Extra	Journal Abbreviation: 2024 IEEE World Forum on Public Safety Technology (WFPST)
Pages	87-92
Proceedings Title	2024 IEEE World Forum on Public Safety Technology (WFPST)
Conference Name	2024 IEEE World Forum on Public Safety Technology (WFPST)
Date Added	20/09/2026, 7:46:50 pm
Modified	20/09/2026, 7:46:50 pm

Tags:

Transportation, security, Wireless networks, Threat modeling, Public security, public safety, Systematics, OTA Update, attacker, Prevention and mitigation, threat model, Web servers

Human-Factor focused Application of STPA to Remotely driven Vehicles

Item Type Conference Paper

Author O. Hans

Author K. Radlak

Author J. Adamy

Abstract This paper presents an analysis of the safety aspects of a Remote Driving System (RDS) using the System-Theoretic Process Analysis (STPA) methodology. The focus is on the human Remote Driver (RD) within the RDS, considering losses and system-level hazards associated with the human-in-the-loop scenario which focuses on the RD actions related to the change in Steering Wheel Angle (SWA). The STPA methodology is applied to identify Unsafe Control Actions (UCA) related to the change in SWA command, resulting in the identification of numerous Causal Factors (CF) that may lead to hazardous scenarios. Several UCAs and CFs are identified that provide information on potential hazards and risks associated with remote driving technology. Based on a selected UCA where the RD does not change the SWA when the road network requires lateral movement, 25 potential loss scenarios were identified and presented in this paper. The results of this work can contribute to the understanding of potential risks and the development of safety mitigation measures in the context of RDS which need to be validated in further research.

Date 14-15 May 2024

DOI 10.1109/WFPST58552.2024.00026

Extra Journal Abbreviation: 2024 IEEE World Forum on Public Safety Technology (WFPST)

Pages 120-125

Proceedings Title 2024 IEEE World Forum on Public Safety Technology (WFPST)

Conference Name 2024 IEEE World Forum on Public Safety Technology (WFPST)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

risk analysis, Hazards, Wheels, Roads, Public security, Behavioral sciences, Prevention and mitigation, human controller, Human in the loop, remote driving, Remote operator, STPA

Identity-Based Cryptography on IoT devices: balancing security and efficiency for modern use cases

Item Type Conference Paper

Author N. O. Gavriilidis

Author S. T. Halkidis

Author A. Mjeda

Author S. Petridou

Abstract Identity-Based Cryptography (IBC) provides a revolutionary approach to public-key cryptography, since it enables message encryption without the need of previously distributed keys. Using identities, e.g., email addresses or serial numbers, as public keys, IBC simplifies key management and reduces the reliance on certificate authorities providing a promising lightweight solution for modern Internet of Things (IoT) use cases. Moreover, leveraging elliptic curves and pairings as cryptographic building blocks, Identity-Based Encryption (IBE) offers strong security with small key sizes, which makes it even more attractive for resource-constrained IoT devices. This paper evaluates the resource requirements, i.e., execution time and memory, of IBE protocols by conducting experiments on mid-class IoT devices, i.e., Raspberry Pi3B+ and Pi4B+. We provide insights regarding the potential of IBC for sensitive data confidentiality in modern use cases, such as remote patient monitoring, drone-based data collection and Vehicle-to-Vehicle (V2V) communication.

Date 15-17 Oct. 2024

DOI 10.1109/FNWF63303.2024.11028814

Extra Journal Abbreviation: 2024 IEEE Future Networks World Forum (FNWF)

ISBN 2770-7679

Pages 273-278

Proceedings Title 2024 IEEE Future Networks World Forum (FNWF)

Conference Name 2024 IEEE Future Networks World Forum (FNWF)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Data privacy, Vehicular ad hoc networks, Internet of Things, Protocols, Encryption, Constrained Devices, Electronic mail, Elliptic curves, Elliptic Curves, Identity-Based Cryptography, Identity-based encryption, Memory management, Patient monitoring

Implemented Vehicular Communication Systems: Overview and Comparative Analysis

Item Type Conference Paper

Author N. Yakusheva

Author A. Proletarsky

Author M. Basarab

Abstract This paper is intended to the applied Vehicle Network Systems, such as Intelligent Transportation Systems (ITS) and Vehicle Ad-Hoc Networks (VANET). The paper brushed up on the history and state-of-the-art of communications for Vehicle Networks. The paper provides the detailed analysis of implemented Vehicle Network architectures and communications in order to realistically assess the scale of deployment and range of applied technologies for the real existing Vehicle Network Systems. Existing ITSs, VANETs and DASs were evaluated and compared in the article.

Date 26-27 Nov. 2019

DOI 10.1109/TELFOR48224.2019.8971275

Extra Journal Abbreviation: 2019 27th Telecommunications Forum (TELFOR)

Pages 1-4

Proceedings Title 2019 27th Telecommunications Forum (TELFOR)

Conference Name 2019 27th Telecommunications Forum (TELFOR)

Date Added 20/09/2026, 7:46:49 pm

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Tags:

Safety, Reliability, Vehicular ad hoc networks, V2X communication, V2V communication, Network architecture, Satellites, ADAS, VANET, Vehicle-to-everything, ITS, Market research, Telecommunications, Network systems, Wireless fidelity

Innovation and I4.0 Management in Connected and Autonomous Automotive Manufacturing

Item Type Conference Paper

Author S. Shah

Author E. N. Ganji

Author O. Mabbott

Author J. Bate

Abstract The paper aims to present the holistic view on the principles and practices of innovation, innovation management and industry 4.0 within connected and autonomous automotive manufacturing environments. It has been seen through literature that innovation management practices enable companies to compete within the autonomous and connected vehicle market and is considered as an emerging and competitive differentiator towards the growth of the product and that of meeting customer demands within the changing markets. Similarly, industry 4.0 has the capabilities to offer new and innovative insights within manufacturing processes, allowing organisations to exchange data across systems and enable them to utilise the information more effectively and efficiently towards future decision-making capabilities. Virtualization of the process and that of the supply ensures more smooth operations between intra-organisations and external partners and hence more advanced data sharing systems are going to be provide the functionality within the global environments. The paper examines the linkage of innovation management and industry 4.0 practices within connected and autonomous vehicles in automotive sector.

Date 17-20 June 2018

DOI 10.1109/ICE.2018.8436262

Extra Journal Abbreviation: 2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)

Pages 1-8

Proceedings Title 2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)

Conference Name 2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC)
Date Added 20/09/2026, 7:46:50 pm
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Tags:

connected and autonomous vehicles, Technological innovation, Automotive engineering, Industries, Companies, I4.0, information technology, Innovation management, Manufacturing

Intelligent Connected Vehicle Intrusion Detection and Mitigation: An Analysis of Explainable AI

Item Type Conference Paper
Author R. Vishnukumar
Author A. Padma
Author M. Ramaiah
Abstract IoT and AI created a Transportation Management System, resulting in the Internet of Vehicles. Intelligent vehicles are combined with contemporary communication technologies (5G) to achieve automated driving and adequate mobility. IoV faces security issues in the next five (5) areas: data safety, V2X communication safety, platform safety, Intermediate Commercial Vehicles (ICV) safety, and intelligent device safety. Numerous types of AI models have been created to reduce the outcome infiltration risks on ICVs. The need to integrate confidence, transparency, and repeatability into the creation of Artificial Intelligence (AI) for the safety of ICV and to deliver harmless transport systems, on the other hand, has led to an increase in explainable AI (XAI). Therefore, the space of this analysis protected the XAI models employed in ICV intrusion detection systems (IDSs), their taxonomies, and available research concerns. The study's findings demonstrate that, despite its relatively recent submission to ICV, XAI is a potential explore area for those looking to increase the net effect of ICVs. The paper also demonstrates that XAI's greater transparency will help it gain acceptance in the vehicle industry.
Date 22-24 Aug. 2024
DOI 10.1109/ICETCI62771.2024.10704143
Extra Journal Abbreviation: 2024 International Conference on Emerging Techniques in Computational Intelligence (ICETCI)
Pages 252-256
Proceedings Title 2024 International Conference on Emerging Techniques in Computational Intelligence (ICETCI)
Conference Name 2024 International Conference on Emerging Techniques in Computational Intelligence (ICETCI)
Date Added 20/09/2026, 7:46:50 pm
Modified 20/09/2026, 7:46:50 pm

Tags:

Safety, Security, Transportation, Intrusion detection, Surveys, Vehicle-to-everything, Connected intelligent car, XAI, Explainable AI, Routing, Real-time systems, Taxonomy

Investigating External Interaction Modality and Design Between Automated Vehicles and Pedestrians at Crossings

Item Type Conference Paper

Author S. Bai

Author D. D. Legge

Author A. Young

Author S. Bao

Author F. Zhou

Abstract In this study, we investigated the effectiveness and user acceptance of three external interaction modalities (i.e., visual, auditory, and visual+auditory) in promoting communications between automated vehicle systems (AVS) and pedestrians at a crosswalk through a large number of combined designs. For this purpose, an online survey was designed and distributed to 68 participants. All participants reported their overall preferences for safety, comfort, trust, ease of understanding, usability, and acceptance towards the systems. Results showed that the visual+auditory interaction modality was the mostly preferred, followed by the visual interaction modality and then the auditory one. We also tested different visual and auditory interaction methods, and found that “Pedestrian silhouette on the front of the vehicle” was the best preferred option while middle-aged participants liked “Chime” much better than young participants though it was overall better preferred than others. Finally, communication between the AVS and pedestrians' phones was not well received due to privacy concerns. These results provided important interface design recommendations in identifying better combination of visual and auditory designs and therefore improving AVS communicating their intention with pedestrians.

Date 19-22 Sept. 2021

DOI 10.1109/ITSC48978.2021.9564867

Extra Journal Abbreviation: 2021 IEEE International Intelligent Transportation Systems Conference (ITSC)

Pages 1691-1696

Proceedings Title 2021 IEEE International Intelligent Transportation Systems Conference (ITSC)

Conference Name 2021 IEEE International Intelligent Transportation Systems Conference (ITSC)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Safety, Privacy, Visualization, Conferences, Animation, Human voice, Usability

IoT Based Human Biometric System Using Alcohol Level Detection for Driver Security

Item Type Conference Paper

Author A. Sinha

Author N. Kumar

Author A. Ticku

Author V. Modi

Author M. Shah

Author K. Shrivastava

Abstract In today's society, driving after intoxication has become a major factor in traffic accidents. It is common knowledge that operating a vehicle while intoxicated is a major contributing factor in a significant number of accidents. The effects of drunk driving go beyond the drunk driver; they also affect other drivers and passengers in the same vehicle. As a result, this growing threat needs to be addressed immediately. Our project's goal is to create a system that uses the Internet of Things (IoT) to find traces of alcohol in someone's breath and apply preventive measures in order to effectively address this issue. Although there are laws against drunk driving, it is difficult to enforce them widely because law enforcement cannot constantly be present on the roads to spot drunk drivers. This restriction increases the chance of accidents significantly. In order to reduce future risks, a trustworthy system that can recognize and discourage drunk drivers is urgently needed. Drunk driving deaths are increasing, so we need to figure out how to reduce these kinds of deaths. The breath analyzing sensor used in the suggested system is the MQ-3, which calculates the breath alcohol content of an individual. The amount of alcohol that is allowed in the blood in India is 0.03. The engine won't start if the person's blood alcohol content is higher than this limit, making it impossible for them to drive. Additionally, the system will send an alert, possibly through a Raspberry Pi, to a friend or relative, providing the exact location using GPS coordinates. This notification serves as a precautionary measure to keep others informed about the situation. Implementing such a system can contribute significantly to reducing accidents related to alcohol consumption. In conclusion, detectors like the one proposed in this project hold immense significance in preventing the loss of life and property caused by reckless driving under the influence of alcohol.

Date 29-30 Dec. 2023

DOI 10.1109/ICAIIHI57871.2023.10489696

Extra Journal Abbreviation: 2023 International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIIHI)

Volume 1

Pages 1-7

Proceedings Title 2023 International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIIHI)

Conference Name 2023 International Conference on Artificial Intelligence for Innovations in Healthcare Industries (ICAIIHI)

Date Added 20/09/2026, 7:46:49 pm

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Tags:

Security, Accidents, Vehicles, User experience, Technological innovation, Internet of Things, Software, Global Positioning System, Alcohol level, Blood, driver security, IoT,HCI, Solids, System

Issues and Challenges in VANET Routing Protocols

Item Type Conference Paper

Author A. Ahamed

Author H. Vakilzadian

Abstract Intelligent transportation systems (ITS) are becoming more popular due to the demand for advanced cyber-physical systems and comfort applications and services needed for use in autonomous vehicles. Vehicular ad hoc Networks (VANETs) are an essential part of ITS, as they support vehicle-to-vehicle and vehicle-to-infrastructure data communication. The performance of a VANET solely depends on the routing protocols in the network layer, which provide efficient and effective data communication. In this paper, we survey the most recent issues surrounding efficient routing protocols that are suitable for VANETs. We also address the pros and cons of these protocols in terms of scalability, quality of services, security and privacy, energy efficiency, transmission bandwidth limitation, and broadcasting issues.

Date 3-5 May 2018

DOI 10.1109/EIT.2018.8500180

Extra Journal Abbreviation: 2018 IEEE International Conference on Electro/Information Technology (EIT)

ISBN 2154-0373

Pages 0723-0728

Proceedings Title 2018 IEEE International Conference on Electro/Information Technology (EIT)

Conference Name 2018 IEEE International Conference on Electro/Information Technology (EIT)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Vehicular ad hoc networks, VANET, Quality of service, Routing, energy efficiency, Scalability, Delays, Routing protocols, broadcasting issues and challenges, quality of services, scalability, security and privacy, transmission bandwidth limitaion

Keep Your Distance: A Cognitive Approach to Mobile Robot-Cyclist Interaction

Item Type Conference Paper

Author Y. Strigina

Author O. Singler-Hack

Author R. M. Kramm

Author J. Ehlers

Author F. Schrödel

Abstract This paper proposes a framework for integrating the subjective safety of vulnerable road users into autonomous driving (AD) systems' decision-making. Cognitive-psychological data on cyclists are collected in laboratory experiments and validated in an empirical field test. The identified subjective safety

parameters ensure the optimal distance and velocity for AD systems, represented by a mobile robot. Approaching decision-making design from a human-centred perspective ensures safety, comfort, and high acceptance of AD systems.

Date 27-28 Oct. 2025

DOI 10.1109/ICVES65691.2025.11376576

Extra Journal Abbreviation: 2025 IEEE International Conference on Vehicular Electronics and Safety (ICVES)

Pages 110-115

Proceedings Title 2025 IEEE International Conference on Vehicular Electronics and Safety (ICVES)

Conference Name 2025 IEEE International Conference on Vehicular Electronics and Safety (ICVES)

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Tags:

Autonomous vehicles, Safety, Decision making, Urban areas, Traffic Safety, Stress, Mobile robots, Human Factors, Testing, Weather, Robot sensing systems, Vehicular and wireless technologies, Autonomous Mobile Robots

Leveraging Commercial Connected Vehicle Data for High-Speed Signalized Intersection Safety: A Quantitative Analysis of Hard-Braking and Crashes

Item Type Conference Paper

Author X. Wei

Author S. Knickerbocker

Author N. Hawkins

Author Z. Hans

Author A. Sharma

Abstract This study leverages commercial connected vehicle (CCV) data to address road safety issues at high-speed signalized intersections in Iowa. The study focuses on assessing road safety using minimal CCV data. The temporal stability analysis of the data concluded that the 8-week data was relatively stable. By testing the correlation between the number of hard braking at different intersections and crash data in 8 weeks of randomly selected CCV data, it was concluded that hard braking and crashes have a high correlation (greater than 0.5). Additionally, the study explores the predictive power of CCV data on crash risk and time distribution throughout the day. The results reflect the importance of real-time insights from CCV data for targeted interventions, highlighting the need for a data-driven approach to enhance overall road safety.

Date 14-15 May 2024

DOI 10.1109/WFPST58552.2024.00028

Extra Journal Abbreviation: 2024 IEEE World Forum on Public Safety Technology (WFPST)

Pages 138-142

Proceedings Title 2024 IEEE World Forum on Public Safety Technology (WFPST)

Conference Name 2024 IEEE World Forum on Public Safety Technology (WFPST)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Road safety, traffic safety, Connected vehicles, Public security, Real-time systems, Statistical analysis, Stability analysis, commercial connected vehicle data, Correlation, crash, hard braking events, intersections

LIDAR Data Filtering Enhancing Techniques Development and Validation by Deep Learning

Item Type Conference Paper

Author G. Mariammal

Author N. Siva Rama Lingham

Author R. M. Jasmine

Author R. Aruna

Author P. Mahalakshmi

Author S. Sundari

Abstract WeatherNet is a revolutionary technology that can handle rain, snow, and the darkness. At its core, WeatherNet uses advanced algorithms as intelligent filters, and seamlessly integrates with LiDAR systems to improve safety and performance. Using real-time weather data, the system ensures continuous adaptation to changing weather conditions, maintaining LiDAR point cloud accuracy for accurate identification even in challenging conditions. Adaptive The filtering mechanism dynamically adjusts parameters based on weather fluctuations, optimizing LiDAR data for reliable travel. This real-time optimization is a game-changer, allowing autonomous vehicles to traverse weather conditions with dented accuracy and unprecedented reliability. Beyond functionality, WeatherNet is far-reaching. The implications for more autonomous cars. It disconnects the connection barriers to acceptance of robust solutions to climate problems, enhancing Trustin's ability to drive independently. Additionally, the integration of WeatherNet into the involved vehicle has broad implications for transportation and logistics, allowing regular operation throughout the year and opening up new possibilities for its use in services and distribution, transportation, and emergency situations. In conclusion, intelligent filtering, dynamic optimization, and increased accuracy enhance WeatherNet's security and demonstrate autonomous vehicles, paving the way for easy operation in all weather conditions and tackling critical road blocks with multiple options.

Date 20-22 June 2025

DOI 10.1109/CONIT65521.2025.11166892

Extra Journal Abbreviation: 2025 5th International Conference on Intelligent Technologies (CONIT)

Pages 1-5

Proceedings Title 2025 5th International Conference on Intelligent Technologies (CONIT)

Conference Name 2025 5th International Conference on Intelligent Technologies (CONIT)

Date Added 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

Autonomous vehicles, Accurate identification, Adaptive filtering mechanism, Advanced algorithms, Challenging conditions, Changing weather conditions, Continuous adaptation, Darkness, Deep learning, Dented accuracy, Dynamic parameter adjustment, Far-reaching implications, Intelligent filters, LiDAR point cloud accuracy, LiDAR systems, Optimizing LiDAR data, Rain, Real-time optimization, Real-time weather data, Reliable travel, Revolutionary technology, Safety and performance, Snow, Unprecedented reliability, Weather fluctuations, WeatherNet, Transportation, Heuristic algorithms, Vehicle dynamics, Real-time systems, Laser radar, Accuracy, Optimization, Meteorology

Multiple intention tracking by a generalized potential field approach

Item Type Conference Paper**Author** F. Particke**Author** M. Hiller**Author** L. Patino-Studencki**Author** C. Sippl**Author** C. Feist**Author** J. Thielecke

Abstract Fully automated vehicles and mobile robots operate in a shared environment with pedestrians. To minimize the risk for pedestrians, it is very important to track them in a precise way. As cameras are often installed in surveillance situations, they are used for tracking pedestrians in a shared environment. To improve the accuracy of the tracking, it is necessary to include all available context information in the fusion process. One important information source is the intention of the pedestrian. A generalized potential field is used, which can be modeled using pedestrian movements. When the intention of the person is unknown, different hypotheses for the intention of the pedestrian are considered. A Multi-Hypotheses tracking filter fuses the intention information and the pedestrian position measurements of a camera, whereby the tracking accuracy is improved. The proposed approach is evaluated using real camera data from a simple scenario in Edinburgh Informatics Forum. All results are evaluated in dependence of the measurement quality and the frame rate of the camera. The Multi-Hypotheses based tracking outperforms the simple Kalman filter over the whole range of frame rates and standard deviations.

Date 10-12 Oct. 2017**DOI** 10.1109/SDF.2017.8126388**Extra** Journal Abbreviation: 2017 Sensor Data Fusion: Trends, Solutions, Applications (SDF)**Pages** 1-5**Proceedings Title** 2017 Sensor Data Fusion: Trends, Solutions, Applications (SDF)**Conference Name** 2017 Sensor Data Fusion: Trends, Solutions, Applications (SDF)

Date Added 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

Acceleration, Kalman filters, Cameras, Trajectory, Informatics, Position measurement, Tracking

On the detection of grey hole and rushing attacks in self-driving vehicular networks

Item Type Conference Paper**Author** K. M. Ali Alheeti**Author** A. Gruebler**Author** K. D. McDonald-Maier

Abstract Vehicular ad hoc networks play an important role in the success of a new class of vehicles, i.e. self-driving and semi self-driving vehicles. These networks provide safety and comfort to passengers, drivers and vehicles themselves. These vehicles depend heavily on external communication to predicate the surrounding environment through the exchange of cooperative awareness messages (CAMs) and control data. VANETs are exposed to many types of attacks such as black hole, grey hole and rushing attacks. In this paper, we present an intelligent Intrusion Detection System (IDS) which relies on anomaly detection to protect external communications from grey hole and rushing attacks. Many researchers agree that grey hole attacks in VANETs are a substantial challenge due to them having their distinct types of behaviour: normal and abnormal. These attacks try to prevent transmission between vehicles and roadside units and have a direct and negative impact on the wide acceptance of this new class of vehicles. The proposed IDS is based on features that have been extracted from a trace file generated in a network simulator. In our paper, we used a feed-forward neural network and a support vector machine for the design of the intelligent IDS. The proposed system uses only significant features extracted from the trace file. Our research, concludes that a reduction in the number of features leads to a higher detection rate and a decrease in false alarms.

Date 24-25 Sept. 2015**DOI** 10.1109/CEEC.2015.7332730**Extra** Journal Abbreviation: 2015 7th Computer Science and Electronic Engineering Conference (CEEC)**Pages** 231-236**Proceedings Title** 2015 7th Computer Science and Electronic Engineering Conference (CEEC)**Conference Name** 2015 7th Computer Science and Electronic Engineering Conference (CEEC)**Date Added** 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

self-driving car, Vehicles, Feature extraction, Intrusion detection, security, intrusion detection system, Roads, Routing protocols, Ad hoc networks, semi self-driving car, vehicular ad hoc networks

On the Security of an IoT-based Intelligent Transportation Service

Item Type Conference Paper

Author Z. Garofalaki

Author D. Kallergis

Abstract Within the Internet of Things (IoT) ecosystem, an intelligent transportation service may incorporate data sharing with third-party services to enhance fast and accurate decision-making. In essence, this IoT-based transportation service may dynamically adapt an extra-domain one and adjusts several service lifecycle and key-processes, such as the fleet management. The depth of this adaptation can be highlighted by possible changes of the service's security. However, the literature seems to lack an IoT-based service security assessment which also considers such IoT features. In this work, we rely on the stochastic modeling method to depict the data flow changes in regard to the fleet management using the Petri net ontology. The fleet management process is evaluated in two customised real-life scenarios to highlight how third-party data flow changes can affect this process lifecycle. Then, we assess the security for the two scenarios to estimate the degree in which the service is affected. The modeling process and the security assessment show that the third-party data type is related to the transportation service and the fleet management key-process in terms of their operation and security.

Date 20-22 Sept. 2019

DOI 10.1109/SEEDA-CECNSM.2019.8908369

Extra Journal Abbreviation: 2019 4th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)

Pages 1-5

Proceedings Title 2019 4th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)

Conference Name 2019 4th South-East Europe Design Automation, Computer Engineering, Computer Networks and Social Media Conference (SEEDA-CECNSM)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Security, Transportation, Decision making, security, IoT, Internet of Things, Data integration, Stochastic processes, intelligent transportation service, Petri nets

P2V and V2P communication for Pedestrian warning on the basis of Autonomous Vehicles

Item Type Conference Paper

Author A. Hussein

Author F. García

Author J. M. Armingol

Author C. Olaverri-Monreal

Abstract The use of smartphones in a road context by drivers and Vulnerable Road Users (VRU) is rapidly increasing. To reduce the risks related to the influence of smartphone usage in a situation where traffic needs to be considered, a collision prediction algorithm is proposed based on Pedestrian to Vehicle (P2V) and Vehicle to Pedestrian (V2P) communication technologies, which increases the visual situational awareness of VRU regarding the nearby location of both autonomous and manually-controlled vehicles in a user-friendly form. The proposed application broadcasts the device's position to the vehicles nearby, and reciprocally, the vehicles nearby broadcast their position to the device in use, supporting pedestrians and other VRU to minimize potential dangers and increase the acceptance of autonomous vehicles on our roads. Results regarding the evaluation of the proposed approach showed a good performance and high detection rate, as well as a high user satisfaction derived from the interaction with the system.

Date 1-4 Nov. 2016

DOI 10.1109/ITSC.2016.7795885

Extra Journal Abbreviation: 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)

ISBN 2153-0017

Pages 2034-2039

Proceedings Title 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)

Conference Name 2016 IEEE 19th International Conference on Intelligent Transportation Systems (ITSC)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Autonomous vehicles, Safety, Prediction algorithms, Smart phones, Roads

Performance Analysis of Ring Oscillator PUF for Robust Security in Smart Transportation

Item Type Conference Paper

Author P. K. Sadhu

Author V. P. Yanambaka

Author A. Abdelgawad

Author K. Yelamarthi

Abstract In smart cities, smart transportation (ST) is one of the critical infrastructures for efficient traffic routing, congestion-free roads, and for development of sustainable transportation. The advancement of vehicular technologies, Internet of Things (IoT), high-speed communication channel, allows ST to become a major application of IoT system and an important area for research. Unlike traditional transport systems, ST demands secure communication and rapid validation for ensuring data integrity. In this context, a 64-bit Physical Unclonable Function (PUF) was developed for use in the ST. In this work, PUF shows 94.14% reliability, 34.02% uniqueness, and the time required to generate key is 95.8ms.

Date 14 June-31 July 2021

DOI 10.1109/WF-IoT51360.2021.9596038

Extra Journal Abbreviation: 2021 IEEE 7th World Forum on Internet of Things (WF-IoT)

Pages 301-302

Proceedings Title 2021 IEEE 7th World Forum on Internet of Things (WF-IoT)

Conference Name 2021 IEEE 7th World Forum on Internet of Things (WF-IoT)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Security, Smart transportation, Smart cities, Internet of Things, Routing, Roads, Physical unclonable function, Physical Unclonable Function, Ring oscillators, Vehicular and wireless technologies

Privacy Preservation in IoT: Anonymization Methods and Best Practices

Item Type Conference Paper

Author M. Vardalachakis

Author M. Tampouratzis

Abstract The Internet of Things (IoT) offers the most intense technological attempt, allowing objects to collect and exchange vast amounts of information efficiently. While this interconnectivity has various advantages, it also brings severe risks to each individual or organization regarding privacy. As the population of connected devices increases, so does the opportunity to abuse sensitive information, which needs reinforced privacy strategies. Risk masking has become one of the most effective strategies for data leaks. This study will examine various anonymization methods, such as homomorphic encryption, data masking, aggregation methods, pseudonymization, and differential privacy, that could be implemented with IoT data. Looking at each method using its performance, pros, and cons offers a comprehensive view of how various methods might be used to protect the user's privacy without compromising the data's utility. In addition to the introduction of multiple methods for anonymization, this study also highlights the privacy preservation methods in IoT environments based on the best practices described. It stresses the importance of determining the value of data, acquiring user acceptance, and following the law, particularly the CCPA and the GDPR. Moreover, it also discusses the importance of building initial privacy in the industry guided by businesses, developers, and users who are shaping the privacy industry's initial norms. Given the problem and solution for ensuring privacy in IoT, this study aims to provide theoretical depth with some practical examples. Powerful anonymization practices can help stakeholders gain more trust from users, increasing the ethical expansion of IoT (in a privacy-focused direction) and good business practices.

Date 20-22 Nov. 2024

DOI 10.1109/CIEES62939.2024.10811203

Extra Journal Abbreviation: 2024 5th International Conference on Communications, Information, Electronic and Energy Systems (CIEES)

Pages 1-6

Proceedings Title 2024 5th International Conference on Communications, Information, Electronic and Energy Systems (CIEES)
Conference Name 2024 5th International Conference on Communications, Information, Electronic and Energy Systems (CIEES)
Date Added 20/09/2026, 7:46:50 pm
Modified 20/09/2026, 7:46:50 pm

Tags:

Privacy, Differential privacy, Internet of Things, Stress, Industries, Protection, aggregation methods, anonymization techniques, Best practices, CCPA compliance, differential privacy, GDPR compliance, homomorphic encryption, Homomorphic encryption, Information filtering, Information integrity, privacy preservation in IoT, pseudonymization, risk masking

Recognition System: Detection of License Plate

Item Type Conference Paper

Author E. I. Singh

Author P. Rahi

Author V. Jain

Author N. Sharma

Author Y. Anand

Author A. K. Shukla

Abstract License Plate Recognition (LPR) systems offer increased efficiency, security, and automation across various domains, making them an essential tool for modern surveillance, management, and law enforcement efforts. In this paper, we present a new approach to LPR systems to solve the image ambiguity problem with high extraction and recognition accuracy. Integrating machine learning models like YOLO (You Only Look Once) V8 for region detection and convolutional neural networks (CNNs) for post-processing into a license plate recognition (LPR) system can significantly enhance its performance and accuracy. Using a dataset of approximately 8000 Indian license plate images for training and testing is a significant step in developing a robust license plate recognition (LPR) system tailored to Indian conditions, with accuracy of 92%. Image denoising as an integral part of the license plate recognition (LPR) system brings a novel dimension to the overall approach. Applying this to machine learning models can significantly improve the performance of LPR systems. The system uses advanced algorithms to correctly identify image components, allowing users to pivot on the relevant parts of the license plate. The utilization of CNNs further refines the extracted data, enabling precise recognition of alphanumeric characters despite image distortions. Following character extraction, the system undergoes a crucial verification process by cross-referencing the extracted data with the Indian Regional Transport Office (RTO) dataset. This verification step ensures the reliability and authenticity of the recognized license plate information. Remarkably, the proposed system achieves a verification accuracy of 96.5%, underscoring its efficacy in real world scenarios. The implications of this research extend beyond academic discourse, offering tangible benefits for modern transportation systems. By enhancing the accuracy and reliability of LPR

systems, particularly in scenarios involving blurred or challenging images, the proposed methodology contributes to improved traffic management, enhanced security measures, and streamlined administrative processes. The ability to accurately identify and authenticate license plates in real-time has numerous practical applications across various sectors, particularly in law enforcement, transportation, and security. Moreover, the proposed system holds promise for integration into various smart city initiatives, where advanced technologies are leveraged to optimize urban infrastructure and enhance the overall quality of life for residents. It will be paid. This system is a comprehensive approach to solving problems caused by ambiguous images. Through the integration of reinvigorated machine learning models and dataset interpretation, the proposed system achieves high accuracy and reliability for practical applications in modern transportation systems and beyond.

Date 8-9 Aug. 2024

DOI 10.1109/ICCPCT61902.2024.10673076

Extra Journal Abbreviation: 2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT)

Volume 1

Pages 171-177

Proceedings Title 2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT)

Conference Name 2024 7th International Conference on Circuit Power and Computing Technologies (ICCPCT)

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Tags:

Law enforcement, Transportation, Machine learning, Accuracy, Training, Denoiser, Image, Image recognition, License, Segmentation, YOLO

Required Trustworthiness Level based on Threat Analysis and Risk Assessment (TARA)

Item Type Conference Paper

Author A. R. Ferraz de Lucena

Author A. Hermann

Author N. Trkulja

Author A. Kiening

Author A. Petrovska

Author F. Kargl

Abstract Automated vehicles communicating with each other or their surroundings are expected to exchange a large amount of data. With that, the trustworthiness of a shared data item concerning its integrity is raised, as well as the trustworthiness of a vehicle component not having been tampered with by an attacker. Traditional security mechanisms, such as misbehavior detection, can help identify some security violations but cannot assess the overall consequences of a range of vehicle attacks. For this purpose, previous work has already introduced the Trust

Assessment Framework, which computes a target entity's Actual Trustworthiness Level (ATL). This paper focuses on the concept of Required Trustworthiness Level (RTL), which represents the numerical thresholds an ATL needs to reach for an entity to be considered trustworthy. We present a risk-based method to calculate the belief component of an RTL based on the well-established and standardized Threat Analysis and Risk Assessment (TARA). We provide an in-vehicle use case to demonstrate our belief calculation method and discuss the impact of using risk ratings.

Date 15-17 Oct. 2024

DOI 10.1109/FNWF63303.2024.11028144

Extra Journal Abbreviation: 2024 IEEE Future Networks World Forum (FNWF)

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Pages 519-526

Proceedings Title 2024 IEEE Future Networks World Forum (FNWF)

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Tags:

Security, Risk management, Belief Calculation, Threat Analysis and Risk Assessment, Trust Assessment, Trust Requirement

RIoTS: Risk Analysis of IoT Supply Chain Threats

Item Type Conference Paper

Author T. Kieras

Author M. J. Farooq

Author Q. Zhu

Abstract Securing the supply chain of information and communications technology (ICT) has recently emerged as a critical concern for national security and integrity. With the proliferation of Internet of Things (IoT) devices and their increasing role in controlling real world infrastructure, there is a need to analyze risks in networked systems beyond established security analyses. Existing methods in literature typically leverage attack and fault trees to analyze malicious activity and its impact. In this paper, we develop RIoTS, a security risk assessment framework borrowing from system reliability theory to incorporate the supply chain. We also analyze the impact of grouping within suppliers that may pose hidden risks to the systems from malicious supply chain actors. The results show that the proposed analysis is able to reveal hidden threats posed to the IoT ecosystem from potential supplier collusion.

Date 2-16 June 2020

DOI 10.1109/WF-IoT48130.2020.9221323

Extra Journal Abbreviation: 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)

Pages 1-6

Proceedings Title 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)

Conference Name 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)

Date Added 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

Security, Risk management, Reliability, Ecosystems, Supply chains, Internet of Things, Attack Tree, Birnbaum importance, Improvement potential, Information and communication technology, Supply Chain

RISEE: A Highly Interactive Naturalistic Driving Trajectories Dataset with Human Subjective Risk Perception and Eye-tracking Information

Item Type Conference Paper**Author** X. Wu**Author** J. Chen**Author** P. Wang**Author** S. Chen**Author** H. Meng**Author** Y. Shen

Abstract In the research and development (R&D) and verification and validation (V&V) phases of autonomous driving decision-making and planning systems, it is necessary to integrate human factors to achieve decision-making and evaluation that align with human cognition. However, most existing datasets primarily focus on vehicle motion states and trajectories, neglecting human-related information. In addition, current naturalistic driving datasets lack sufficient safety-critical scenarios while simulated datasets suffer from low authenticity. To address these issues, this paper constructs the Risk-Informed Subjective Evaluation and Eye-tracking (RISEE) dataset which specially contains human subjective evaluations and eye-tracking data apart from regular naturalistic driving trajectories. By leveraging the complementary advantages of drone-based (high realism and extensive scenario coverage) and simulation-based (high safety and reproducibility) data collection methods, we first conduct drone-based traffic video recording at a highway ramp merging area. After that, the manually selected highly interactive scenarios are reconstructed in simulation software, and drivers' first-person view (FPV) videos are generated, which are then viewed and evaluated by recruited participants. During the video viewing process, participants' eye-tracking data is collected. After data processing and filtering, 3567 valid subjective risk ratings from 101 participants across 179 scenarios are retained, along with 2045 qualified eye-tracking data segments. The collected data and examples of the generated FPV videos are available in our website https://ivtest-lab.github.io/RISEE_dataset/.

Date 18-21 Nov. 2025**DOI** 10.1109/ITSC60802.2025.11423845**Extra** Journal Abbreviation: 2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)**ISBN** 2153-0017**Pages** 2315-2322

Proceedings Title 2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)
Conference Name 2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)
Date Added 20/09/2026, 7:46:50 pm
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Tags:

Autonomous vehicles, Vehicles, Research and development, Decision making, Data collection, Topology, Merging, Trajectory, Gaze tracking, Videos

Secure mmWave C-V2X Communications Using Cooperative Jamming

Item Type Conference Paper

Author M. Yang

Author Y. Ju

Author L. Liu

Author Q. Pei

Author K. Yu

Author J. J. P. C. Rodrigues

Abstract A lack of well-designed security solutions within the millimeter-Wave (mmWave) cellular vehicle-to-everything (V2X) communications significantly impedes the development of applications within the intelligent transportation system. Cooperative jamming is envisioned as a potential technology that can enhance physical layer security performance for plane networks by selectively choosing jammers from the perspective of the legitimate receiver. We propose a blockage-and-power-based jammer selection strategy to address potential security pitfalls in a mmWave cellular V2X network. With the help of jammers whose interference power falls within the acceptance range of legitimate receivers, transmission confidentiality is secured simultaneously without escalating the instability of connections caused by the time-varying nature of V2X networks. We derive the theoretical expression of secrecy outage probability and secrecy throughput based on our preliminary analysis of association probability from the stochastic geometry approach. Numerical results demonstrate that the proposed secure transmission scheme outperforms other cooperative jamming schemes in terms of secrecy throughput.

Date 4-8 Dec. 2022

DOI 10.1109/GLOBECOM48099.2022.10001684

Extra Journal Abbreviation: GLOBECOM 2022 - 2022 IEEE Global Communications Conference

ISBN 2576-6813

Pages 2686-2691

Proceedings Title GLOBECOM 2022 - 2022 IEEE Global Communications Conference

Conference Name GLOBECOM 2022 - 2022 IEEE Global Communications Conference

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Security, Receivers, Stability analysis, Throughput, Cellular vehicle-to-everything, cooperative jamming, jammer selection, millimeter-Wave, physical layer security, Physical layer security, Power system reliability, Probability

Security challenges in Internet of Vehicles (IoV) environment

Item Type Conference Paper

Author N. Sharma

Author N. Chauhan

Author N. Chand

Abstract With recent advancements in technology, different devices that are used in our day-to-day lives are able to connect to internet, communicate and exchange messages with each other. This new paradigm is called “Internet of Things” and is providing services such as smart hospitals, smart city, home automation etc. “Internet of Things” has changed conventional small scale Vehicular Ad-hoc Network (VANET) to a highly scalable, manageable internet based “Internet of Vehicle” (IoV). IoV is a vehicular network model consisting of vehicles, users and other smart devices connected to network and aims to provide various safety as well as entertainment services. Vehicles in IoV system are equipped with different sensors that collect different types of data and send it to computation unit for computation and analysis, based on which directions and other information is sent to each vehicle. We present a model overview of IoV system in this paper. Security in IoV becomes of paramount importance as any system failure directly affects user safety. In this paper, we discuss security issues, various security attacks and their countermeasures from IoV perspective. We also propose an authentication mechanism for Vehicle-to-Infrastructure (V2I) communication in IoV.

Date 15-17 Dec. 2018

DOI 10.1109/ICSCCC.2018.8703272

Extra Journal Abbreviation: 2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC)

Pages 203-207

Proceedings Title 2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC)

Conference Name 2018 First International Conference on Secure Cyber Computing and Communication (ICSCCC)

Date Added 20/09/2026, 7:46:49 pm

Modified 20/09/2026, 7:46:49 pm

Tags:

Safety, Security, V2V communication, Intelligent Transportation System, Internet of Things, Connected Vehicles, Sensors, Cloud computing, Internet of Vehicles, Computer hacking

Security Considerations on 5G-Enabled Back-Situation Awareness for CCAM

Item Type Conference Paper

Author M. Centenaro

Author S. Berlato

Author R. Carbone

Author G. Burzio

Author G. F. Cordella

Author S. Ranise

Author R. Riggio

Abstract The increasing demand for cooperative, connected, and automated mobility (CCAM) services should proceed at the same pace with the enforcement of security mechanisms that would make CCAM services secure. The first contribution of this paper resides in a review of the ongoing regulatory and standardization activities related to cybersecurity of autonomous vehicles. Then, referring to the ongoing piloting activities funded by the European Union, we focus on the security threats for back-situation awareness (BSA), i.e., a safety-related CCAM service dealing with emergency scenarios. We propose a practical strong authentication method for BSA, and extensively discuss how existing standards can mitigate the security threats of this prominent CCAM service.

Date 10-12 Sept. 2020

DOI 10.1109/5GWF49715.2020.9221064

Extra Journal Abbreviation: 2020 IEEE 3rd 5G World Forum (5GWF)

Pages 245-250

Proceedings Title 2020 IEEE 3rd 5G World Forum (5GWF)

Conference Name 2020 IEEE 3rd 5G World Forum (5GWF)

Date Added 20/09/2026, 7:46:50 pm

Modified 20/09/2026, 7:46:50 pm

Tags:

Autonomous vehicles, Authentication, Europe, Standards, 5G mobile communication

Sim-to-Real Autonomous Vehicle Lane Keeping using Vision

Item Type Conference Paper

Author J. Kocić

Author N. Jovičić

Abstract This paper presents the analysis of the J-Net deep neural network used for autonomous vehicle lane keeping, and its verification in simulated and real conditions. The transition from the simulated world on the personal computer to the real world in laboratory conditions is tackled. In the presented solution, autonomous lane keeping is achieved by analyzing information from visual sensors using a deep neural network. J-Net was developed with an aim to be implemented on an autonomous vehicle platform with limited hardware performance in terms of computing power and memory capacity. Verification of autonomous driving using J-Net was achieved in simulated conditions, using an

open-source simulator for autonomous driving, and in real-world conditions. For the verification in real-world conditions, an autonomous driving system was designed and implemented in the Laboratory of Electronics at the School of Electrical Engineering, University of Belgrade.

Date 23-24 Nov. 2021

DOI 10.1109/TELFOR52709.2021.9653302

Extra Journal Abbreviation: 2021 29th Telecommunications Forum (TELFOR)

Pages 1-8

Proceedings Title 2021 29th Telecommunications Forum (TELFOR)

Conference Name 2021 29th Telecommunications Forum (TELFOR)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Deep learning, machine learning, Computer architecture, Visualization, autonomous driving, Sensors, Hardware, deep learning, vision, Roads, Telecommunications, camera, deep neural network, end-to-end learning, robo-vehicle, sim-to-real, simulator

Smart city mobility: Investigation of RFID adoption within transportation management

Item Type Conference Paper

Author M. N. Alrafaei

Author A. Bushager

Abstract Smart cities era requires changes in the life style of the citizens technically and technologically. Mobility is one of the main domains in smart cities, hence, transportation is the area of focus in this research. Embedding Radio Frequency Identification RFID tags into all vehicles driving in the Kingdom of Bahrain allows tracking and controlling traffic in addition to reducing it at the same time. The study aims at investigating the readiness and acceptance in Bahrain to establish RFID traffic control and enforce it upon all drivers, while measuring the level of awareness of security risk, privacy violation, and their alignment with legislation and policies in the country. Considering all aspects that might affect the decision of the citizen, the study focuses on understanding their willingness to use it, and which parts of the implementation is more acceptable than others. The results of the study show that the citizens are ready and accept the adoption of the technology to reduce and manage traffic, however, they were not quite concerned about the security and privacy issues related to this new technology. Further study is required in the future to spread awareness of the importance of information security and privacy while using new technologies.

Date 22-23 April 2018

DOI 10.1049/cp.2018.1403

Extra Journal Abbreviation: Smart Cities Symposium 2018

Pages 1-6

Proceedings Title Smart Cities Symposium 2018

Conference Name Smart Cities Symposium 2018

Date Added 20/09/2026, 7:46:49 pm**Modified** 20/09/2026, 7:46:49 pm**Tags:**

Awareness, Mobility, RFID tags, Security Risk, Smart City, Traffic Control

State of the art and trends of Vehicle Communication: Overview

Item Type Conference Paper**Author** N. Yakusheva**Author** A. Proletarsky**Author** M. Basarab**Abstract** This paper provides the state-of-the-art of vehicle communications. The current trends of the vehicle communication are defined in the article. Vehicle communication tends to use Internet of Things (IoT), We are discussing the issues of developing Internet of Vehicles (IoV) and consider it as a part of Smart City. The overview provides the detailed analysis of vehicle communications for IoV in order to determine the new trends.**Date** 26-27 Nov. 2019**DOI** 10.1109/TELFOR48224.2019.8971189**Extra** Journal Abbreviation: 2019 27th Telecommunications Forum (TELFOR)**Pages** 1-4**Proceedings Title** 2019 27th Telecommunications Forum (TELFOR)**Conference Name** 2019 27th Telecommunications Forum (TELFOR)**Date Added** 20/09/2026, 7:46:50 pm**Modified** 20/09/2026, 7:46:50 pm**Tags:**

Vehicular ad hoc networks, V2X communication, Smart city, IoT, Smart cities, VANET, Vehicle-to-everything, ITS, Market research, Real-time systems, Protocols, Internet of Vehicles, Next generation networking, Cellular networks, Internet of Vehicles (IoV), Telecommunication network reliability

Technological Opportunities Discovery for Safety Through Topic Modeling and Opinion Mining in the Fourth Industrial Revolution: The Case of Artificial Intelligence

Item Type Journal Article**Author** I. Park**Author** B. Yoon**Author** S. Kim**Author** H. Seol

Abstract In this article, the fourth industrial revolution (4th IR) has features that are distinct from the previous three industrial revolutions in terms of a paradigm shift. Although 4th IR relevant technologies may have positive impacts in many fields, they may also create harmful impacts, just as every previous industrial revolution has. Most studies dealing with the 4th IR have focused on its positive impacts; the present research concentrates on its negative impacts and suggests technological opportunities to hedge against expected risks in the future. To this end, future societal topics are first identified through latent Dirichlet allocation based topic modeling using textual information of futuristic data, which is retrieved from the website, including information on future-related insights. Second, the risk topics related to the future are selected with the help of a weighting score using predefined risk keywords and sentiment analysis. Finally, safe technological opportunities are identified based on textual information derived from patent data related to risk topics with the perspectives on risk analysis that have prior (prediction, prevention) and posterior perspectives (response, recovery). To illustrate technology relevant to the 4th IR, artificial intelligence (AI) is selected. As a result, 36 topics related to future society and six future risk-related topics are suggested that reflects security-relevant properties of AI and three types of threats. Opportunities to ensure or safety through technology are also proposed in four perspectives. Technological opportunities related to safety are explored in two perspectives. One is to expand existing threats, and the other is to suggest new threats. This research can be utilized to plan future safety technology by taking a data-driven approach under the huge paradigm shift of the 4th IR and proposing potential technological solutions that are not future societal scenarios and policy solutions suggested in the previous studies.

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DOI 10.1109/TEM.2019.2928366

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Tags:

Artificial intelligence, risk analysis, Risk analysis, Artificial intelligence (AI), topic modeling, Data models, Resource management, Analytical models, Fourth Industrial Revolution, fourth industrial revolution (4th IR), future risk-related topic, latent Dirichlet allocation (LDA), Risk management, technological opportunities

Testing the Validity of Multiparticipant Distributed Simulation for Understanding and Modeling Road User Interaction

Item Type Journal Article

Author A. H. Kalantari

Author Y. -S. Lin

Author A. Mohammadi

Author N. Merat

Author G. Markkula

Abstract Understanding driver–pedestrian interactions at unsignalized locations has gained additional importance due to recent advancements in vehicle automation. Naturalistic observations can only provide correlational data of limited value for understanding and modeling the mechanisms underlying road user interaction. Therefore, controlled studies in virtual reality (VR) are an important complement, but conventional methods can only accommodate a single human participant. Recently, there has been some interest in studying interactions in VR, by means of distributed simulation, involving multiple human participants. However, there is a lack of validation of this method. Here, we provide a validation study, focusing on a distributed vehicle–pedestrian interaction setup, where pairs of one driver and one pedestrian interacted under various kinematic conditions in a connected virtual environment. To test the validity of the distributed simulation, we used a naturalistic dataset collected in the same U.K. city, at similar locations, and compared the observed behavior between the two settings. Our results indicate a good relative validity of the simulator study, where road users showed similar nonverbal communication behavior in both datasets. As an additional means of validation, we also leveraged a set of game theoretic models that were developed based on the simulator studies, and found that when applied to the naturalistic dataset, we obtained similar (although not identical) model selection results. The findings suggest that distributed simulation can also be useful for development of computational models of interaction. Overall, the findings suggest that distributed simulation can be a highly valuable tool for studying and modeling road user interactions.

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DOI 10.1109/THMS.2025.3591506

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Pages 865-875

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Tags:

Safety, Vehicles, Automation, Decision making, Pedestrians, Human computer interaction, Urban areas, human–computer interaction, decision making, Mathematical models, Behavioral sciences, Ethics, mathematical models, vehicle driving

The Challenges and Opportunities of Human-Centered AI for Trustworthy Robots and Autonomous Systems

Item Type Journal Article

Author H. He

Author J. Gray
Author A. Cangelosi
Author Q. Meng
Author T. M. McGinnity
Author J. Mehnert

Abstract The trustworthiness of robots and autonomous systems (RASs) has taken a prominent position on the way toward full autonomy. This work is the first to systematically explore the key facets of human-centered AI (HAI) for trustworthy RAS (TRAS). We identified five key properties of a TRAS, i.e., RAS must be: 1) safe in any uncertain and dynamic environment; 2) secure, i.e., protect itself from cyber threats; 3) healthy and fault tolerant; 4) trusted and easy to use to enable effective human-machine interaction (HMI); and 5) compliant with the law and ethical expectations. While the applications of RAS have mainly focused on performance and productivity, not enough scientific attention has been paid to the risks posed by advanced artificial intelligence (AI) in RAS. We analytically examine the challenges of implementing TRAS with respect to the five key properties and explore the role and roadmap of AI technologies in ensuring the trustworthiness of RAS in respect of safety, security, health, HMI, and ethics. A new acceptance model of RAS is provided as a framework for HAI requirements and for implementing TRAS by design. This approach promotes human-level intelligence to augment human capabilities and focuses on contribution to humanity.

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DOI 10.1109/TCDS.2021.3132282

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Tags:

Safety, Artificial intelligence, cybersecurity, safety, Human-robot interaction, Robots, Sensors, Computer security, Ethics, Acceptance model, Autonomous systems, human-centered artificial intelligence (HAI), human-robot interaction (HRI), performance of robots and autonomous system (RAS), Robot sensing systems, system health, Trust computing, trustiness, trustworthiness of RAS, worthiness

The Dark Side of AI: Large Language Models as Tools for Cyber Attacks on Vehicle Systems

Item Type Conference Paper

Author Y. Usman

Author P. K. Gyawali

Author S. Gyawali

Author R. Chataut

Abstract The rapid evolution of autonomous vehicles (AVs) presents significant opportunities for enhancing transportation safety and efficiency. However, with increasing connectivity and complex electronic systems, AVs also become vulnerable to cyberattacks. This paper investigates cybersecurity challenges in the realm of AVs, highlighting the role of artificial intelligence (AI), specifically Large Language Models (LLMs), in exploiting vulnerabilities. We analyze various attack vectors, including Controller Area Network (CAN) manipulation, Bluetooth vulnerabilities, and Key Fob hacking, emphasizing the need for proactive cybersecurity measures. Recent incidents, such as the remote compromise of various vehicle models, underscore the urgent need for robust security solutions in the automotive industry. By leveraging LLMs, attackers can craft sophisticated cyber threats targeting AVs, posing risks to both safety and privacy. We introduce HackerGPT, a customized LLM tailored for cyber attack generation, and demonstrate attacks on virtual CAN networks, Bluetooth systems, and Key Fobs. At the same time, our experiments reveal successful compromises in certain vehicle models; limitations exist, particularly in vehicles with advanced encryption and robust signal transmission protocols. However, this research underscores the broader need for increased awareness and proactive security measures in the automotive sector. Our findings aim to contribute significantly to the ongoing discourse on automotive cybersecurity, offering actionable insights for manufacturers and cybersecurity professionals to safeguard the future of mobility.

Date 17-19 Oct. 2024

DOI 10.1109/UEMCON62879.2024.10754676

Extra Journal Abbreviation: 2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)

Pages 169-175

Proceedings Title 2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)

Conference Name 2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Safety, Cybersecurity, Artificial Intelligence, Autonomous Vehicle, Bluetooth, Cyberattacks, Large Language Model, Automotive engineering, Large language models, Cyberattack, Industries, Protocols, Ethics, Training, Vectors

The Future of Smart Roads: Technology, Challenges and Potential

Item Type Conference Paper

Author A. Badshah

Author A. Daud

Author T. Alsahfi

Author M. Ramzan

Author S. Raza

Author T. Saleem

Abstract The rapid vehicle growth and the resulting traffic congestion have significantly intensified safety concerns. Road accidents cause around 1.19 million deaths each year, with an additional 20 to 50 million people injured. Furthermore, congestion results in the loss of 50 billion hours each year, with an additional 3 billion gallons of fuel wasted. These critical challenges highlight the urgent need for solutions that can enhance road safety, efficiency, and sustainability. This study investigates Emerging Computing Technologies (ECT) integrated into smart roads as potential solutions. Key technologies (e.g., advanced communication systems, the Internet of Things (IoT), sensors, Artificial Intelligence (AI), big data, blockchain, and energy harvesting) are explored for their ability to address these challenges. By reviewing the applications, we examine how smart roads can provide sustainable solutions for transportation systems, including smart traffic lights, road damage detection and maintenance, intelligent street lighting, and energy-efficient infrastructure. Additionally, we discuss the challenges associated with smart roads, including technical, financial, regulatory, cyber-security, privacy concerns, and social acceptance. Finally, we look ahead to the future of smart roads by exploring advancements in emerging communication technologies, AI, blockchain, and energy harvesting solutions that could revolutionize the efficiency, safety, and sustainability of road networks.

Date 23-25 April 2025

DOI 10.1109/ICETECC65365.2025.11071111

Extra Journal Abbreviation: 2025 International Conference on Emerging Technologies in Electronics, Computing, and Communication (ICETECC)

Pages 1-6

Proceedings Title 2025 International Conference on Emerging Technologies in Electronics, Computing, and Communication (ICETECC)

Conference Name 2025 International Conference on Emerging Technologies in Electronics, Computing, and Communication (ICETECC)

Date Added 20/09/2026, 7:46:50 pm

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Tags:

Autonomous vehicles, Safety, Artificial intelligence, Road safety, Sustainable development, Autonomous Vehicle, Intelligent Transportation System, Big Data, Internet of Things, Energy harvesting, Smart roads, Urban Mobility, Communication systems, Blockchains

The Impact of Gender, Seniority, Knowledge, and Interest on Attitudes to Artificial Intelligence

Item Type Journal Article

Author A. Kovačević

Author E. Demić

Abstract Artificial intelligence (AI) has become deeply rooted in our lives, yet uncertainties persist regarding public attitudes to it, particularly among young individuals poised to engage with AI in their future careers. Understanding their perspectives is crucial not only for shaping educational frameworks, but also for assessing students' readiness to navigate the rapidly evolving technological landscape in the modern workspace. This paper examines students' attitudes to AI, as well as their interest in and knowledge about it. An adapted version of the Pew Research Center survey was used in our study to explore how gender and student seniority influence attitudes to AI generally and in specific applications such as facial recognition and self-driving cars. Our aim was to test the effects of these factors on AI attitudes, and to discover how various factors such as socio-demographics, knowledge, and interest may individually or collectively impact on AI attitudes in general, as well as in specific areas such as self-driving cars, facial recognition, or social media algorithms for fake news. We also investigated whether knowledge of AI and interest in it may serve to predict attitudes beyond the impacts of student seniority and gender. Our findings indicate that males self-report greater interest than females, but similar knowledge and general attitudes as female participants. Senior students possessed more AI knowledge compared to freshmen, but similar attitudes towards AI in general and self-driving cars. Interest in AI emerged as a significant predictor of general attitudes to AI and to self-driving cars, suggesting that increased interest correlates with more positive attitudes.

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Journal Abbr IEEE Access

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Tags:

Artificial intelligence, Education, Emotion recognition, education, Psychology, Behavioral sciences, Gender issues, technological innovation

The Peril of Artificial Intelligence

Item Type Conference Paper

Author A. Mathew

Abstract The integration of AI technology is with the hope of reducing human errors, improving efficiency, and providing faster result. Nonetheless, research indicates that AI may become hazardous over time and would not perform the intended roles. To recognize the perilous effects of AI, the study draws from the dynamic programming theory developed by Richard Bellman. The study proves that while AI is programmed to do a useful task, a malfunction may lead to negative consequences of the entire system. It may also attract indirect consequences such as traffic jams. The best way to explain this correlation is via the dynamic programming theory, which asserts that as systems become complex, they may sway from their targeted objective. Computer-related dangers may emanate from

the cognitive complexity of the system as opposed to the specific properties of which the system was meant to perform. The immediate risks of AI are autonomous weapons, although other long-term effects such as gradual replacement of humans by robots and AI drug addiction will also result.

Date 8-10 Jan. 2020

DOI 10.1109/ICISC47916.2020.9171226

Extra Journal Abbreviation: 2020 Fourth International Conference on Inventive Systems and Control (ICISC)

Pages 919-924

Proceedings Title 2020 Fourth International Conference on Inventive Systems and Control (ICISC)

Conference Name 2020 Fourth International Conference on Inventive Systems and Control (ICISC)

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Tags:

Artificial intelligence, AI, Complexity theory, Dynamic programming, Weapons, Correlation, AI weapons, autonomous weapons, Control systems, Drugs, Risks of AI

Toward Robust 3D Perception for Autonomous Vehicles: A Review of Adversarial Attacks and Countermeasures

Item Type Journal Article

Author K. T. Yasas Mahima

Author A. G. Perera

Author S. Anavatti

Author M. Garratt

Abstract At present the perception system of autonomous vehicles is grounded on 3D vision technologies along with deep learning to process depth information. Although deep learning models for 3D perception give promising results, recent research demonstrates that they are also vulnerable to adversarial attacks similar to deep learning models trained on 2D images. As a result, it is essential to further explore the vulnerabilities of 3D perception models in autonomous vehicles and find methods to cope with the risks associated with these adversarial vulnerabilities, in order to improve the social acceptance of commercial autonomous vehicles. This study aims to provide an in-depth overview of the recent adversarial attacks and countermeasures against 3D perception models on autonomous vehicles. Further, challenges associated with the research domain and future research directions are highlighted to make autonomous vehicles robust against adversarial attacks.

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autonomous vehicles, Autonomous vehicles, Security, LiDAR, Adversarial machine learning, 3D perception, Reviews, Sensors, deep learning, Solid modeling, Adversarial attacks, Three-dimensional displays

Towards Corner Case Detection for Autonomous Driving

Item Type Conference Paper

Author J. -A. Bolte

Author A. Bar

Author D. Lipinski

Author T. Fingscheidt

Abstract The progress in autonomous driving is also due to the increased availability of vast amounts of training data for the underlying machine learning approaches. Machine learning systems are generally known to lack robustness, e.g., if the training data did rarely or not at all cover critical situations. The challenging task of corner case detection in video, which is also somehow related to unusual event or anomaly detection, aims at detecting these unusual situations, which could become critical, and to communicate this to the autonomous driving system (online use case). Such a system, however, could be also used in offline mode to screen vast amounts of data and select only the relevant situations for storing and (re)training machine learning algorithms. So far, the approaches for corner case detection have been limited to videos recorded from a fixed camera, mostly for security surveillance. In this paper, we provide a formal definition of a corner case and propose a system framework for both the online and the offline use case that can handle video signals from front cameras of a naturally moving vehicle and can output a corner case score.

Date 9-12 June 2019

DOI 10.1109/IVS.2019.8813817

Extra Journal Abbreviation: 2019 IEEE Intelligent Vehicles Symposium (IV)

ISBN 2642-7214

Pages 438-445

Proceedings Title 2019 IEEE Intelligent Vehicles Symposium (IV)

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Tags:

Autonomous vehicles, Security, Object recognition, Machine learning, Cameras, Surveillance, Robustness, Videos, Semantic segmentation, Training data

Trustworthy and Efficient P2P Updates in Drones and Autonomous Aircrafts to Ensure Public Safety

Item Type Conference Paper

Author N. M. I. Chowdhury

Author S. N. Zisad

Author R. Hasan

Abstract In public safety applications, drones and autonomous aircraft are frequently deployed for mission-critical tasks. Ensuring the integrity and trustworthiness of their operational software-particularly during updates-is essential to maintaining system reliability and safety. Traditional centralized update mechanisms present significant vulnerabilities, including inefficiency, susceptibility to single points of failure, and security breaches, rendering them inadequate for such highstakes environments. To address these challenges, we propose a decentralized, peer-to-peer (P2P) update framework that incorporates a Merkle hash tree-based protocol for integrity verification. This approach not only ensures the authenticity and integrity of software updates but also enhances system resilience by eliminating reliance on centralized infrastructure. We conduct a comprehensive evaluation of our method, comparing it against conventional centralized update architectures. Our findings demonstrate that the proposed solution offers superior scalability, robustness, and security. This research contributes a novel, decentralized update mechanism tailored for autonomous systems, underpinned by cryptographic integrity guarantees provided by the Merkle tree structure.

Date 23-25 Sept. 2025

DOI 10.1109/WF-PST65083.2025.00028

Extra Journal Abbreviation: 2025 IEEE World Forum on Public Safety Technology (WF-PST)

Pages 143-148

Proceedings Title 2025 IEEE World Forum on Public Safety Technology (WF-PST)

Conference Name 2025 IEEE World Forum on Public Safety Technology (WF-PST)

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Tags:

Drones, security, Robustness, Public security, Protocols, Scalability, Software, Aircraft, drone, P2P, Peer-to-peer computing, Rendering (computer graphics), Trees (botanical), update, verification

Understanding Citizens' Emotional Pulse in a Smart City Using Artificial Intelligence

Item Type Journal Article

Author A. Adikari

Author D. Alahakoon

Abstract Over the past decade, smart city applications have gained significant attention in industrial informatics. However, little attention has been given to perceiving the emotions and perceptions of citizens who have a direct impact on smart city initiatives. In this article, we propose the use of publicly available abundant social media conversations that contain contextual information encompassing citizens' emotions and perceptions, which could be considered to provide the means to feel the “emotional pulse” of a city. We propose an automated AI-based observation framework to detect the emergence of public emotions and negativity in conversations. We evaluated the applicability of the framework using 29 928 social media conversations toward the much-debated topic of self-driving vehicles which will become increasingly relevant to smart cities. The patterns and transitions of citizens' collective emotions were modeled using the Natural Language Processing and Markov models while the negativity (toxicity) in conversations was evaluated using a deep learning based classifier. The framework could be adopted by industry leaders and government officials for smart observation of citizen opinions to improve security, communication, and policymaking.

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Tags:

Artificial intelligence, Deep learning, Natural Language Processing (NLP), smart city, Smart cities, Twitter, Computational modeling, emotions modeling, Informatics, Markov models

Understanding User Perspectives: Exploring Product Services, Development, and Sentiment Analysis

Item Type Conference Paper

Author G. B. Jagnade

Author K. Kumbhare

Author R. Agrawal

Author C. Dhule

Abstract The automotive industry's dynamic landscape is increasingly shaped by customer satisfaction, preferences, and the quality of auto care services. This research endeavours to unravel the intricacies of customer experiences through one-to-one calls and sentiment analysis. A comprehensive dataset, encompassing diverse demographic variables and detailed ownership attributes, was analysed to discern patterns and insights. Our findings reveal multifaceted aspects of automotive ownership, customer expectations, and the role of auto care services. Demographic information, such as gender, occupation, and vehicle details, has

been meticulously examined. Moreover, sentiment analysis was employed to gauge customer sentiment regarding their vehicles, auto care, and service providers. The study unveils distinctive clusters of car owners based on their attitudes and expectations. We also scrutinise the factors that drive vehicle expenses and influence budgeting decisions. The implications of our research are far-reaching, shedding light on the evolving landscape of automotive ownership and customer satisfaction. We discuss how auto care services can better align with the preferences and expectations of car owners. This study contributes to the broader understanding of customer-centric approaches within the automotive industry and lays the foundation for further research in this domain.

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Extra Journal Abbreviation: 2024 International Conference on Innovations and Challenges in Emerging Technologies (ICICET)

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Tags:

Sentiment analysis, Customer satisfaction, Recurrent neural networks, Industries, Data models, Accuracy, Training, Automotive Industry Auto Care Services, Customer Satisfaction, Enhancement Strategies, Machine Learning Algorithms

V2X Message Multi-Label Classification: A Deep Learning Approach

Item Type Conference Paper

Author C. F. Lazaga

Author M. A. Rosales

Author G. P. Tan

Abstract This is a simulation study created to provide insights into using deep learning models to improve Vehicle-To-Everything (V2X) communications security by classifying the types of messages transmitted or received. The dataset is provided in the IEEE Data Port, pre-processed by one-hot encoding, tokenizer, post-padding, and embedding layer. After text preprocessing, the dense vectorize values are inputted into these four deep learning algorithms: Bidirectional Recurrent Neural Network (BRNN), Convolutional – Bidirectional Recurrent Neural Network (CNN-BRNN), Bidirectional Long Short-Term Memory (BLSTM), and Convolutional Bidirectional Long Short-Term Memory (CNN-BLSTM). Researchers use the TensorFlow and Keras python libraries using their default parameters of these models. A flatten and dense layer with a Softmax activation function is also used in its output layer for multilabel classification. The results of the classification metrics observed that the CNN-BRNN produced a superior performance than the rest of the models using the V2X Dataset, having an accuracy of 90%, precision of 98.46%, recall of 80.12%, f1-score of 84.49%,

and loss value of 0.45. It has also been observed that BRNN has a better performance than BLSTM models with or without Convolutional Layers.

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DOI 10.1109/ICoAIT67446.2025.11308890

Extra Journal Abbreviation: 2025 1st International Conference on Artificial Intelligence Technology (ICoAIT)

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Tags:

Security, V2X, Deep learning, Convolutional neural networks, Vehicle-to-everything, Recurrent neural networks, Measurement, Accuracy, Bidirectional Long Short-Term Memory, Bidirectional Recurrent Neural Networks, Convolutional – Bidirectional Long Short-Term Memory, Convolutional – Bidirectional Recurrent Neural Networks, Deep Learning Models, Libraries, Long short term memory, Multi label classification

Validation of Assurance Case for Dynamic Systems

Item Type Conference Paper

Author U. D. Ferrell

Author A. H. A. Anderegg

Abstract Aviation safety has accrued and applied decades of understanding on known risks and effective mitigations. That knowledge captured in compliance standards - can be tested for predictable outcomes. Autonomy that involves learning systems tend to be dynamic and may continuously be adapting to their environment. Such continuous adaptation has inherent unknown risks depending upon the guardrails imposed on the learning systems. Design assurance and use of traditional standards are inadequate for these dynamic systems. Assurance Cases are used to present an argument for the assurance of systems. Dynamic systems require that assurance cases be continuously validated. One method of validation is using real time collection of Safety Performance Indicators (SPIs) which are crafted during the development of the system. This paper presents the need for SPIs and methods for creating and nurturing the SPIs to help all stakeholders. This method shadows regulations and allows risk-based approvals that may be applied for both conventional and for novel technology. Aviation is facing enormous growth in autonomous technology, reuse of components of unknown pedigree, and new aircraft designs that do not fit into Type classification. Mitigations are more heavily connected to operations, training, and other components of the ecosystem itself. The challenge is to make an assurance case for vehicles within the ecosystem. The automobile industry, which is similarly challenged by dynamically changing autonomous systems, is finding some possible solutions to build safer systems. UL 4600, a Standard for Safety for the Evaluation of Autonomous Products applies to fully autonomous road vehicles. The goal-based, technology-neutral features of UL 4600 have been extended to apply to aviation.

So applied, the assurance process is adaptable to innovation and discovery while encouraging the current practices of standards compliance and taking a System of Systems (SoS) view. It proposes an assurance case that is an organized argument that a system is acceptable for its intended use with respect to specified concerns (such as safety, security, correctness). This paper gives guidance for validation of an assurance case through monitoring SPI within the operational context. The method by monitoring safety performance indices in the operational environment provides continued validation even as the ecosystem, components and controls change. For approval of novel systems including UAS and AAM, with features that do not lend themselves to traditional compliance methods, regulators have embraced the Safety Continuum perspective, which focuses on safety performance achieving expected outcomes. The performance-based assurance methods can be used with initially wider performance margins for certification of novel products, components of unknown pedigree, and autonomous vehicles. As the performance range is better known the margins can be decreased. Further, this paper recognizes that a one-time initial approval/acceptance is not adequate for learning systems and novel features. The continued validation through performance supports fast-paced development and product evolution. The initial assurance case for a product can limit risk through a closed environment until the margin for some unknowns is validated. For example, if the performance of collision avoidance function using new technology is not known, larger alert limits may be implemented until more confidence is gained after validating the assurance case via SPIs. The approach of monitored SPI throughout the life of the product is now feasible with the aid of big data processing. The aviation industry is already using similar methods for identifying maintenance problems. As systems grow more autonomous, more machine-to-machine exchanges are involved, making it easy to extend the monitoring and prediction practices to SPI. The method also allows for variants and derivatives of the baseline to have their own assurance case within the context of the baseline argument. The key is replacing design approval with through-life assurance that connects continuous operational safety into both the design and airworthiness determinations. The determination is predicated on the monitored SPIs and predicted performance of the product remaining consistent with the assurance argument predictions. This enables even complex automated products to be audited for airworthiness with an evolving ecosystem based on monitored and predictive data. Another advantage of the performance-based assurance case is the public comprehensibility of safety. With SPIs and predictions of performance the automobile segment has paved the way for public scrutiny of automated vehicles. The use of SPIs in aeronautical product assurance will facilitate transparency. This could be accomplished through appropriate dashboards to aid public perception and explain events and precautions taken during the evolution toward more autonomous aviation vehicles. This could reflect a stepwise evolution of complexity. This paper explores how the aviation industry can apply performance-based assurance case methods to assure new and novel as well as systems of unknown pedigree. The same framework could then be extended to autonomous systems and new types of aircraft which do not fit the current Type classification. One of the major benefits of this technology agnostic method are faster risk-based approvals of novel technology within a Safety Continuum.

Date 18-22 Sept. 2022

DOI 10.1109/DASC55683.2022.9925731

Extra Journal Abbreviation: 2022 IEEE/AIAA 41st Digital Avionics Systems Conference (DASC)

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Tags:

Safety, AI, Ecosystems, Learning systems, UAM, Autonomous Systems, Predictive analytics, Machine Learning, Industries, Regulators, Certification, AAM, Complexity, Dynamic Systems, Dynamical systems, Epistemic Safety, Goal-based Regulatory Oversight, Safety Performance Based Assurance Case, Safety Performance Indicators, System-of-Systems, UAS, Uncertainty in Performance

What is Stopping Digital Agricultural Production Technologies from Farmers

Item Type Conference Paper
Author E. Vuka
Author Y. Wu
Abstract With the increasing challenges in agriculture and the pressing need for food security, from the evidence of our previous work and other researchers' investigation, digital agricultural technologies are promising sustainable agricultural production and reducing environmental impact. However, the adoption of digital agriculture is still lagging behind and leaves a research gap. To gain insights into the barriers to the uptake of these technologies, we conducted a content analysis of relevant literature. Our analysis innovatively introduced a “Three Parties” concept that simplifies the barriers to adoption by examining the interplay between farmers, technology, and external factors. We identified several key barriers, including education level, age, and risk perception of farmers, as well as cyber and data security, and access threshold of technology itself. External policy and market influences were also found to be critical factors. Our findings highlighted the most important risks from the technology itself are stability, reliability or accuracy, and access. We also suggested the need for further research to overcome these barriers and accelerate the adoption of digital technologies in farming.
Date 19-21 Nov. 2024
DOI 10.1109/CINTI63048.2024.10830897
Extra Journal Abbreviation: 2024 IEEE 24th International Symposium on Computational Intelligence and Informatics (CINTI)
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Proceedings Title 2024 IEEE 24th International Symposium on Computational Intelligence and Informatics (CINTI)
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Tags:

Reliability, security, Education, Data security, Reviews, agriculture, technology, Production, Accuracy, Content management, digital, Food security, production, Smart agriculture, Stability analysis, sustainable

“What's Happening”- A Human-centered Multimodal Interpreter Explaining the Actions of Autonomous Vehicles

Item Type Conference Paper

Author X. Luo

Author F. Ding

Author R. Chen

Author R. Panda

Author J. Loo

Author S. Zhang

Abstract Public distrust of self-driving cars is growing. Studies emphasize the need for interpreting the behavior of these vehicles to passengers to promote trust in autonomous systems. Interpreters can enhance trust by improving trans-parency and reducing perceived risk. However, current so-lutions often lack a human-centric approach to integrating multimodal interpretations. This paper introduces a novel Human-centered Multimodal Interpreter (HMI) system that leverages human preferences to provide visual, textual, and auditory feedback. The system combines a visual interface with Bird's Eye View (BEV), map, and text display, along with voice interaction using a fine-tuned large language model (LLM). Our user study, involving diverse par-ticipants, demonstrated that the HMI system significantly boosts passenger trust in AVs, increasing average trust lev-els by over 8%, with trust in ordinary environments rising by up to 30%. These results underscore the potential of the HMI system to improve the acceptance and reliability of autonomous vehicles by providing clear, real-time, and context-sensitive explanations of vehicle actions.

Date 28 Feb.-4 March 2025

DOI 10.1109/WACVW65960.2025.00129

Extra Journal Abbreviation: 2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)

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Pages 1073-1080

Proceedings Title 2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)

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Tags:

Visualization, Computer vision, Large language models, Autonomous automobiles, Real-time systems, Conferences, Prompt engineering, Reliability engineering

Work-in-Progress: Crash Course: Can (Under Attack) Autonomous Driving Beat Human Drivers?

Item Type Conference Paper

Author F. Marchiori

Author A. Brighente

Author M. Conti

Abstract Autonomous driving is a research direction that has gained enormous traction in the last few years thanks to advancements in Artificial Intelligence (AI). Depending on the level of independence from the human driver, several studies show that Autonomous Vehicles (AVs) can reduce the number of on-road crashes and decrease overall fuel emissions by improving efficiency. However, security research on this topic is mixed and presents some gaps. On one hand, these studies often neglect the intrinsic vulnerabilities of AI algorithms, which are known to compromise the security of these systems. On the other, the most prevalent attacks towards AI rely on unrealistic assumptions, such as access to the model parameters or the training dataset. As such, it is unclear if autonomous driving can still claim several advantages over human driving in real-world applications. This paper evaluates the inherent risks in autonomous driving by examining the current landscape of AV sand establishing a pragmatic threat model. Through our analysis, we develop specific claims highlighting the delicate balance between the advantages of AVs and potential security challenges in real-world scenarios. Our evaluation serves as a foundation for providing essential takeaway messages, guiding both researchers and practitioners at various stages of the automation pipeline. In doing so, we contribute valuable insights to advance the discourse on the security and viability of autonomous driving in real-world applications.

Date 8-12 July 2024

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Extra Journal Abbreviation: 2024 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)

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Tags:

Autonomous vehicles, Security, Artificial intelligence, Automation, Threat modeling, Training, Fuels